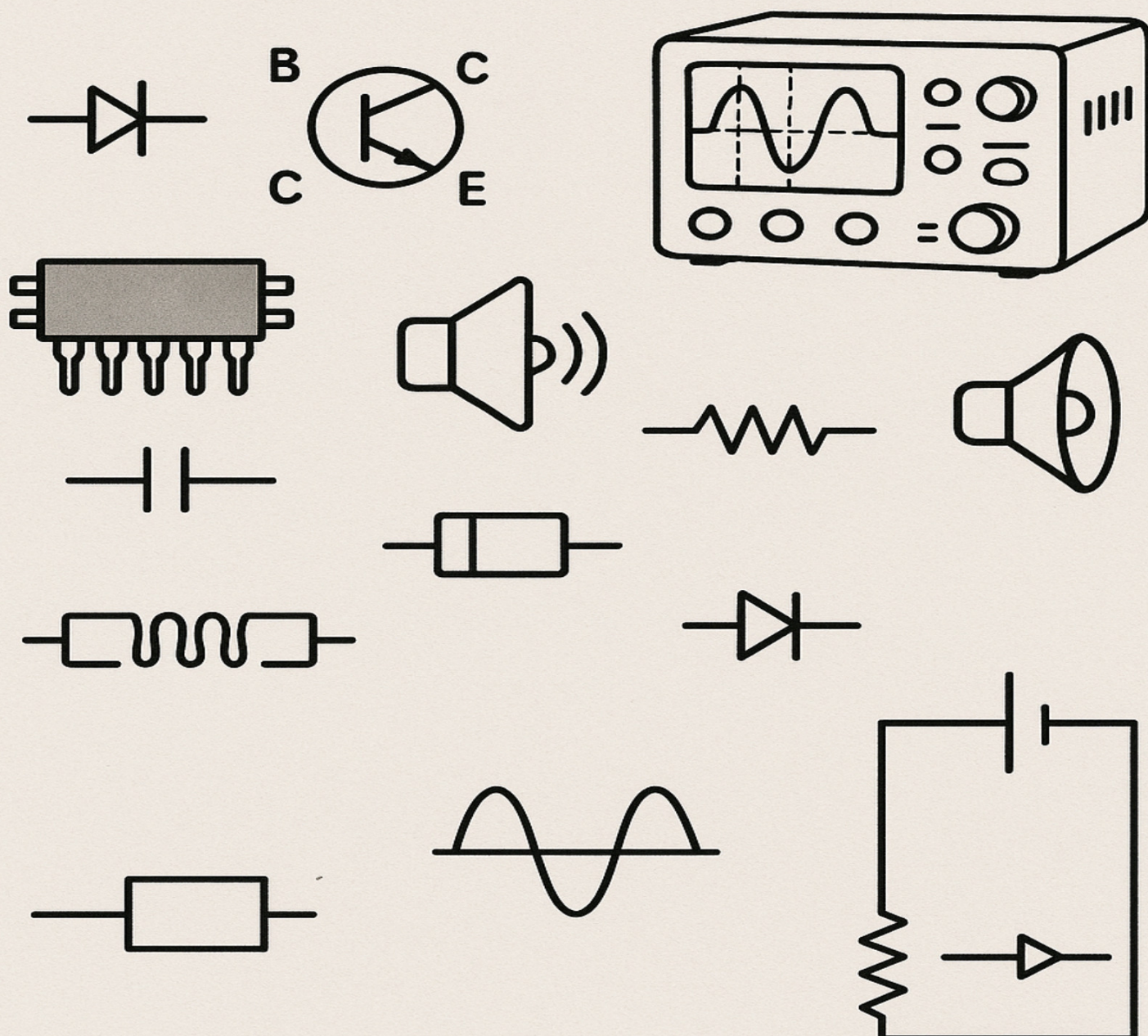
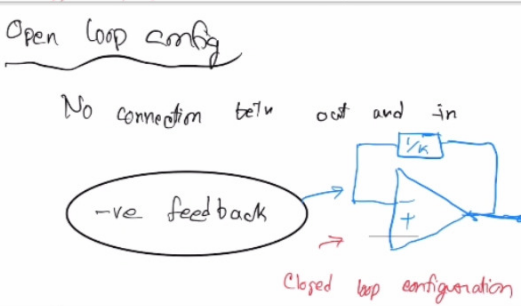
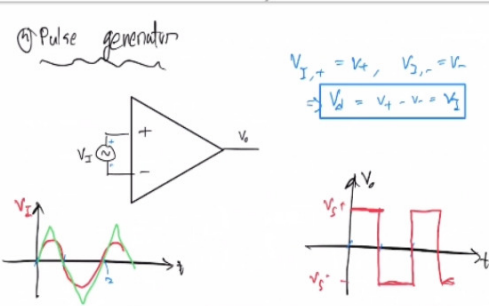
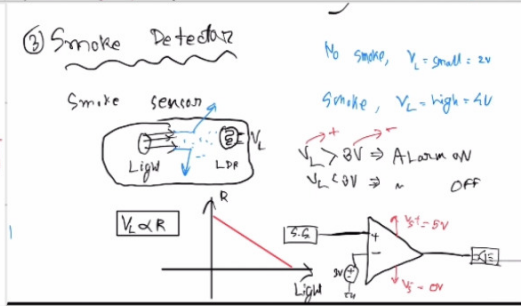
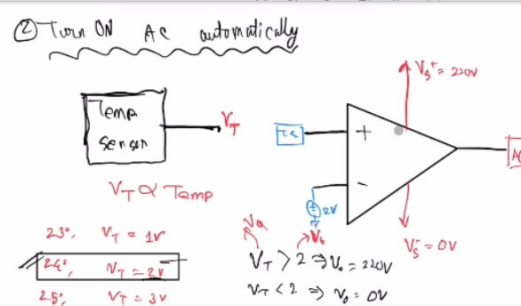
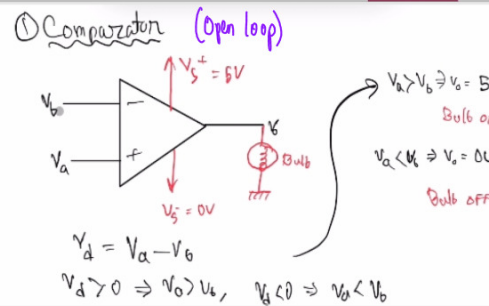
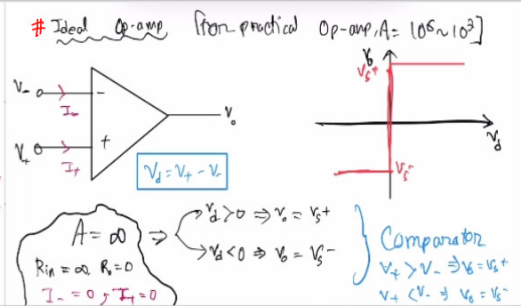
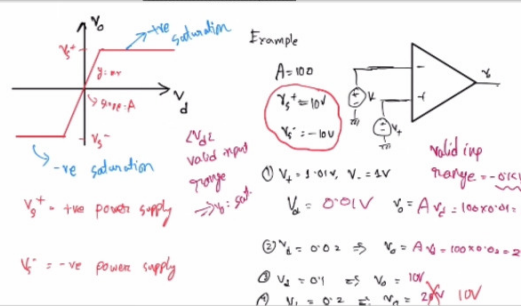
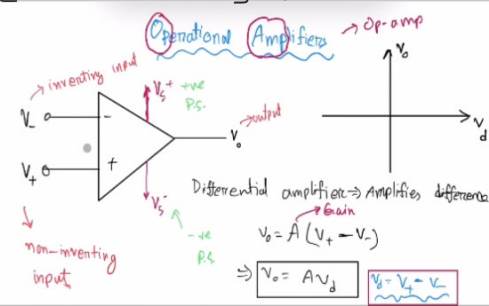


CSE 251

ELECTRONIC DEVICES AND CIRCUITS

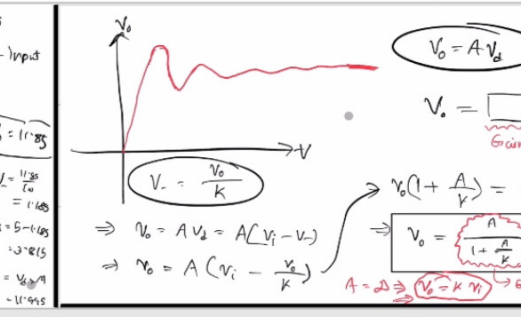
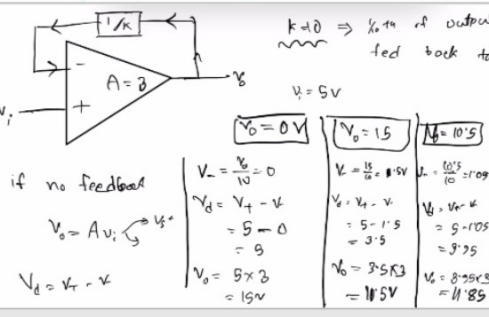




Little tie in input results in V_s^+ . Same for -ve

Output is unstable, main reason is that ideal opamp gain is ∞ .

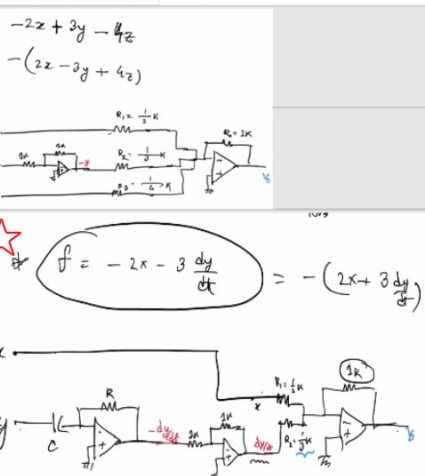
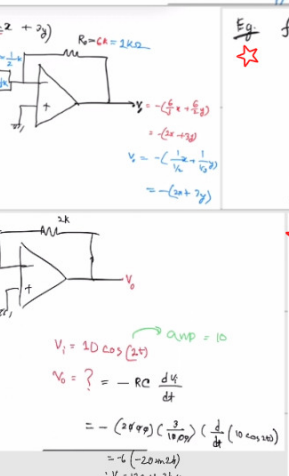
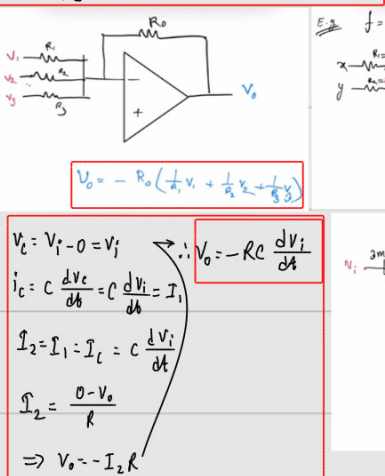
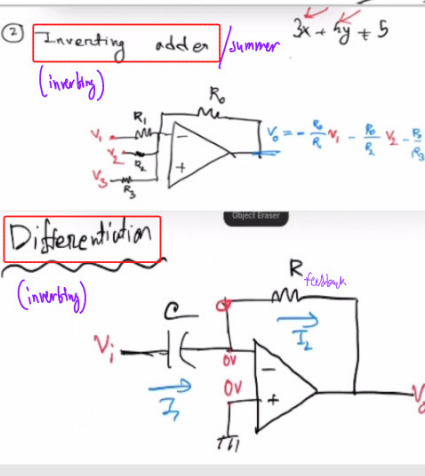
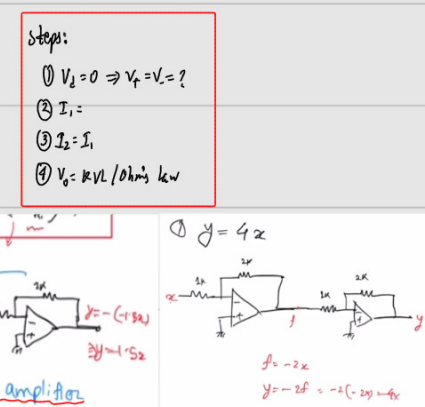
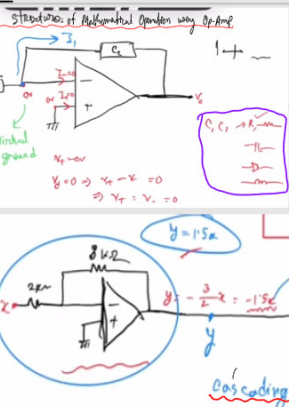
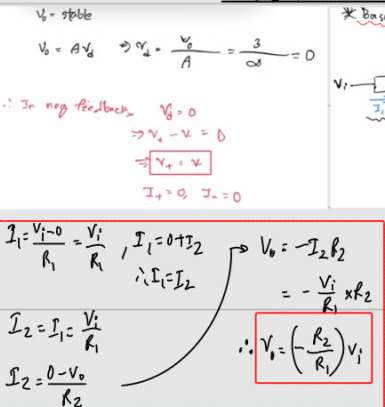
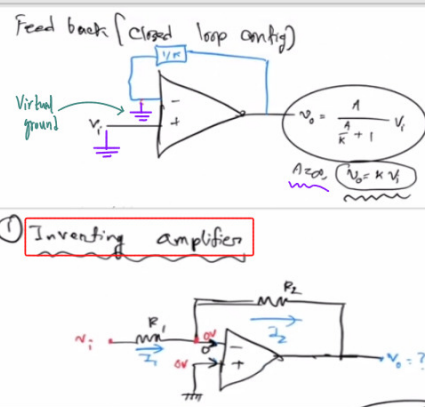
Thus for proper control $\rightarrow$ -ve Feedback



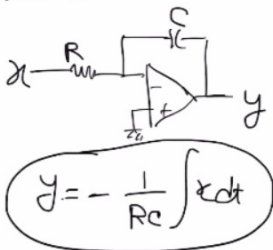
Little tie in input results in V_s^+ . Same for -ve

Output is unstable, main reason is that ideal opamp gain is ∞ .

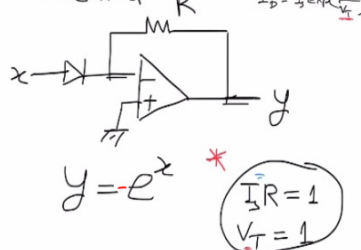
Thus for proper control $\rightarrow$ -ve Feedback



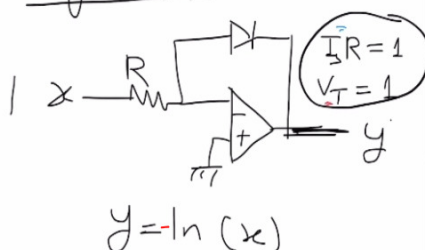
② Integrator



* Exponential



* Logarithm

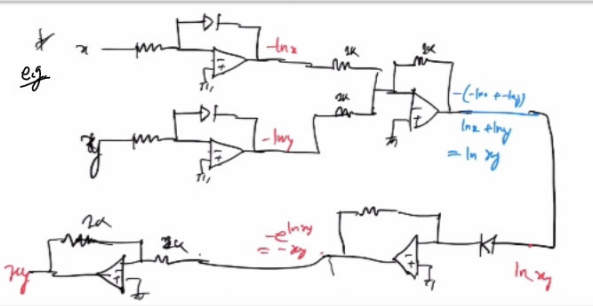


① $\ln(\exp(a)) = a$, $\frac{e^{\ln x}}{\ln} = x$

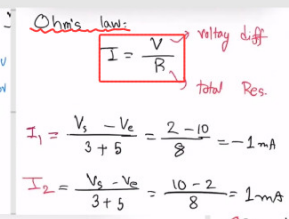
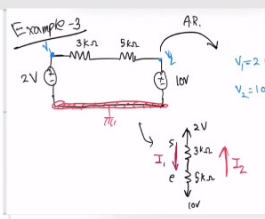
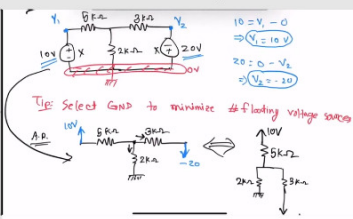
② $\ln(ab) = \ln a + \ln b$

$f = xy$

$$\left. \begin{array}{l} x \rightarrow \ln x \\ y \rightarrow \ln y \end{array} \right\} \ln x + \ln y \rightarrow \ln xy \xrightarrow{\exp} xy$$



Line Diagram



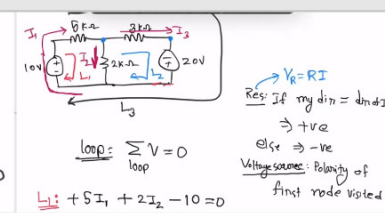
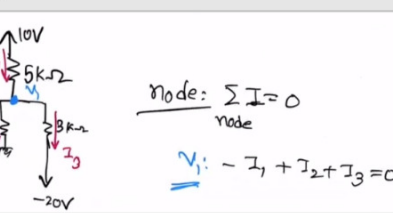
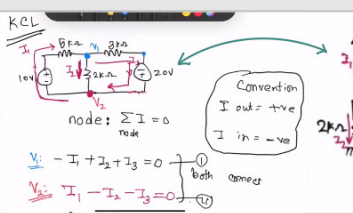
AC / Large electrical devices use
1/2 A current limit
Pharos/BJT/MOSFET/Complex
electronic devices use mA current
Currents in A (instead of mA) will
make these devices to burn.

Mili-Kilo convention

$$mA \times K = V$$

$$10^{-3} \times 10^3 = 1$$

KCL, KVL (Mesh & Nodal Analysis)



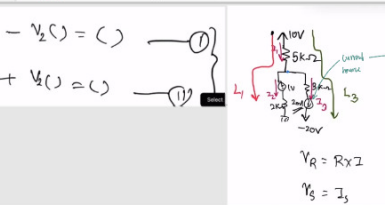
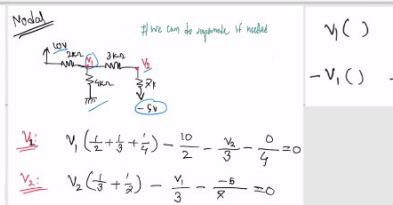
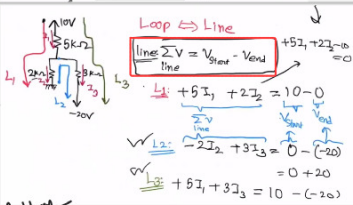
Loop: $\sum V = 0$

$$L_1: +5I_1 + 2I_2 - 10 = 0$$

$$L_2: +3I_3 - 20 - 2I_2 = 0$$

$$L_3: +5I_1 + 3I_3 - 20 - 10 = 0$$

$$L_3 = L_1 + L_2$$



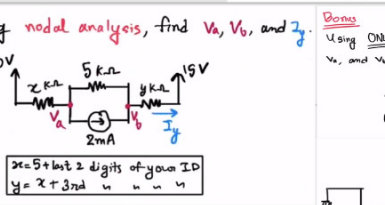
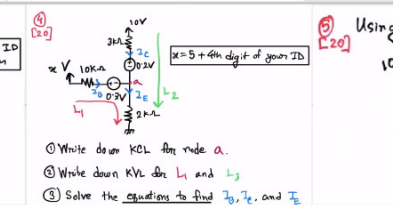
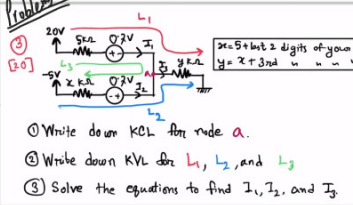
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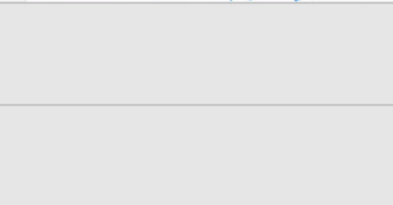
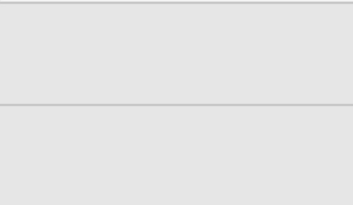
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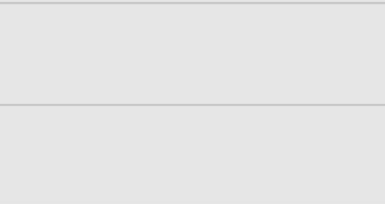
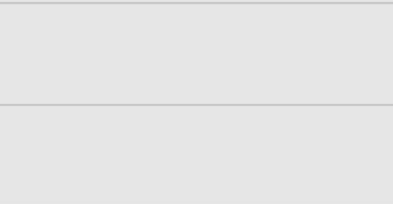
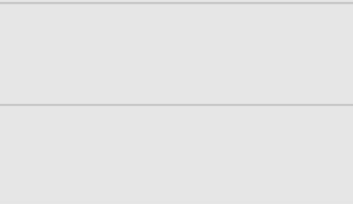
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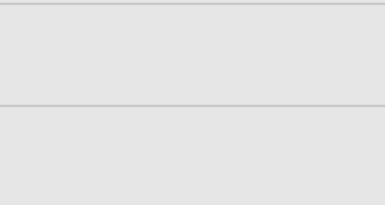
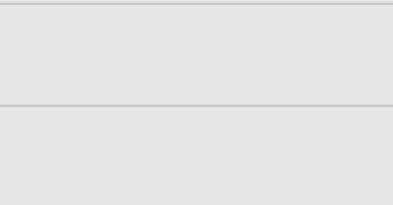
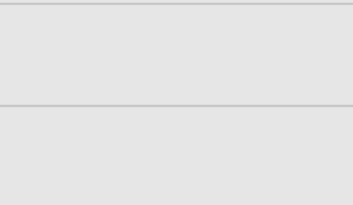
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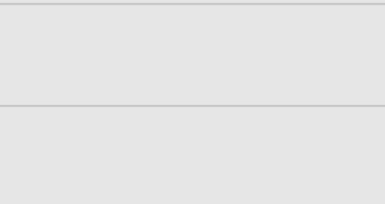
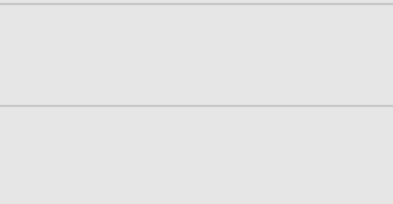
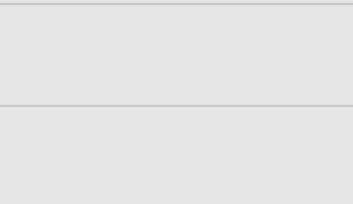
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Loop: $\sum V = 0$

$$L_1: +5I_1 + 2I_2 - 10 = 0$$

$$L_2: +3I_3 - 20 - 2I_2 = 0$$

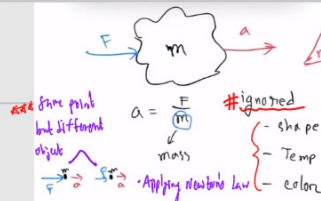
$$L_3: +5I_1 + 3I_3 - 20 - 10 = 0$$

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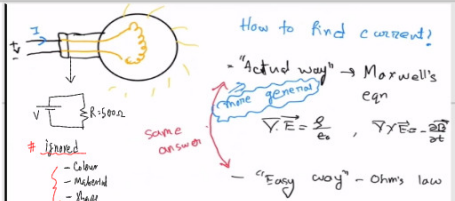
Topics

- Lumped Matter Abstraction $\rightarrow$ Making things easier
- IV characteristics
- IV char. of simple linear devices
- IV char. of compound linear devices
- Equivalent circuit and IV char

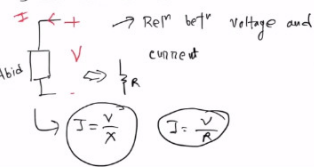
Point-mass Abstraction



Lumped Matter Abstraction



I-V Characterization



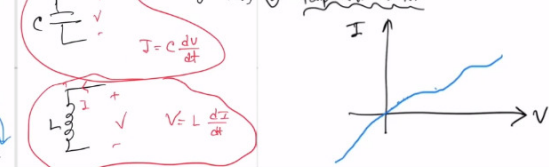
Tabular form

V (V)	I (mA)
1	0.5
2	1
-2	-1
3	1.5

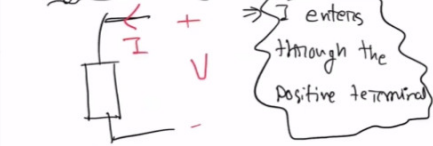
Functional form

Functional form: $I = f(V)$ or $V = g(I)$.
 $f(x) = ?$
 $I = \frac{1}{R} V$
 $y = f(x) = \frac{x}{R}$

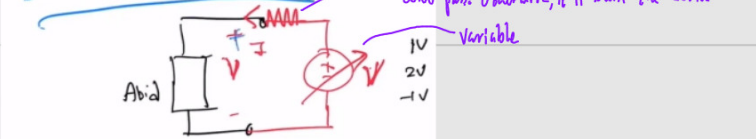
Graphical form



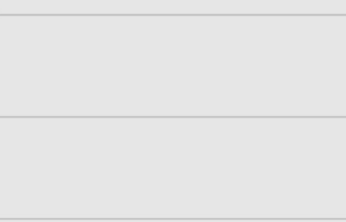
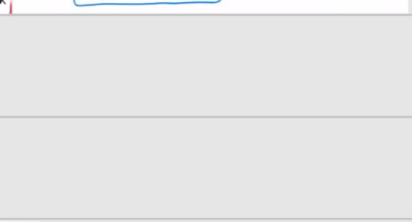
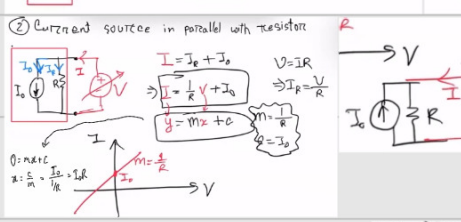
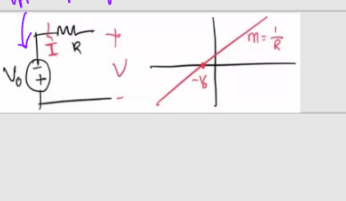
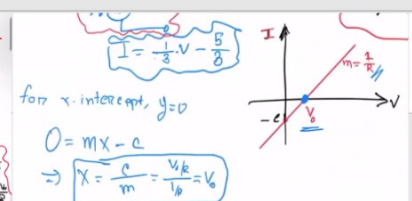
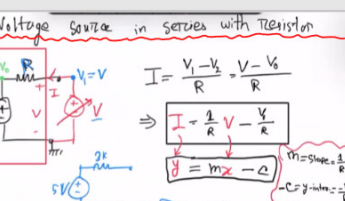
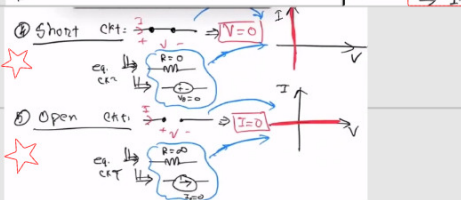
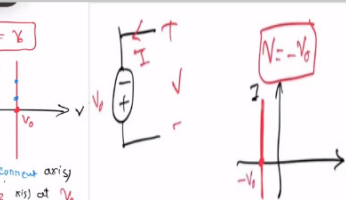
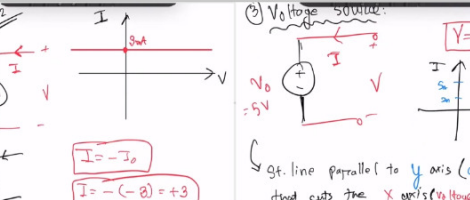
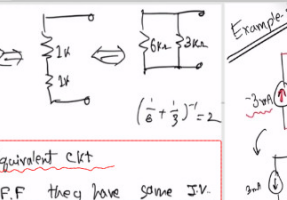
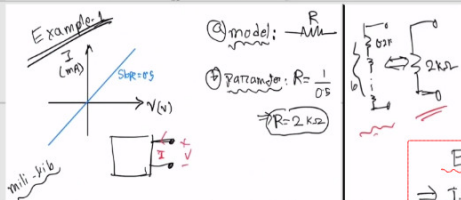
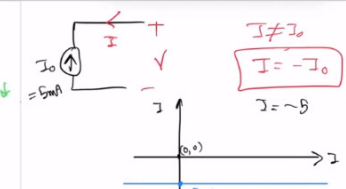
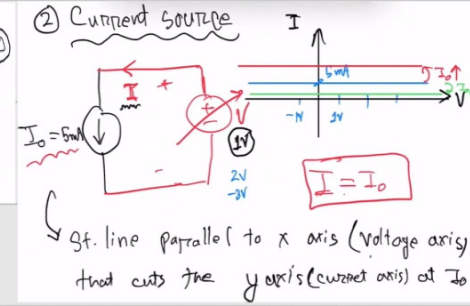
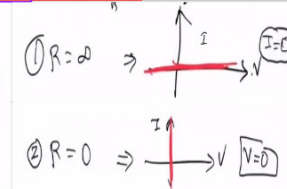
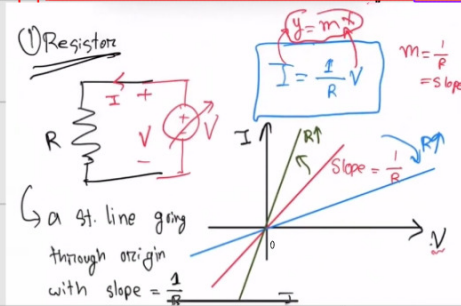
Sign convention



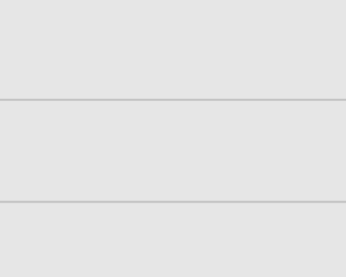
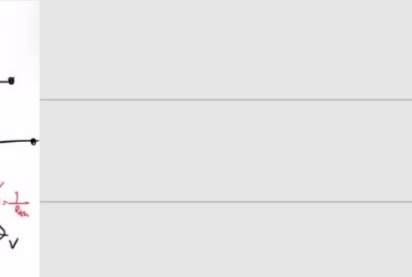
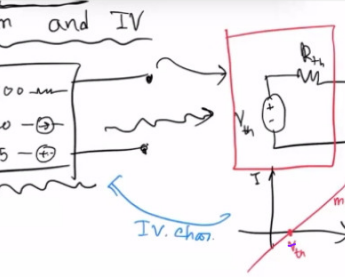
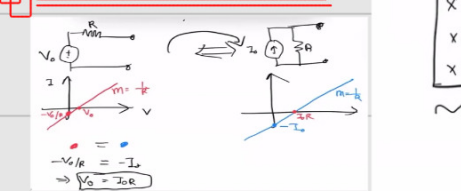
How to find IV



IV Characteristics of Simple Linear devices

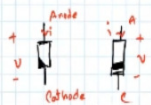
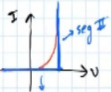


Source Transformation & IV

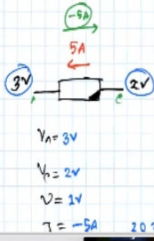


Introduction to Biode

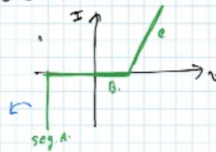
Element non-linear Device $\rightarrow$ piecewise linear



$$V = V_A - V_C$$



modeling non-linear devices using linear



<u>condition</u>	<u>model</u>	<u>parameter</u>
1. The data is stationary	AR	ϕ
2. The data is stationary	MA	θ
3. The data is stationary	ARMA	ϕ, θ
4. The data is non-stationary	ARIMA	ϕ, θ, d
5. The data is non-stationary	ARIMA	ϕ, θ, d, k

- [illegible]

$\rightarrow$ 100-ohm resistor
 $V_{100} = 10V$
 $\Rightarrow I = 0.2mA$

for $R_3 = 3k, I_{R3} = 0.2mA$
 $\Rightarrow V_{R3} = 0.2mA \times 3k = 0.6V$
 $\Rightarrow I = 0.2mA$

what is the current when $V_1 = 2V$?

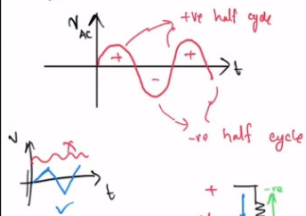
$2V$

will operate in reg. $\Rightarrow$

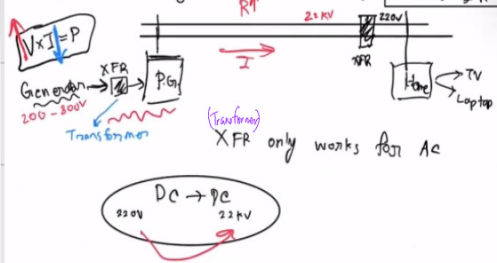
$I_{R3} = 0.2mA$
 $\Rightarrow I = 0.5V + 2$
 $\Rightarrow I = 0.5 \times 2 + 2 = 1mA$

② $I = 1 \text{ mA}$, $V = ?$
 $\hookrightarrow \text{seg 2} \Rightarrow V = 0$

Diode AC $\rightarrow$ DC



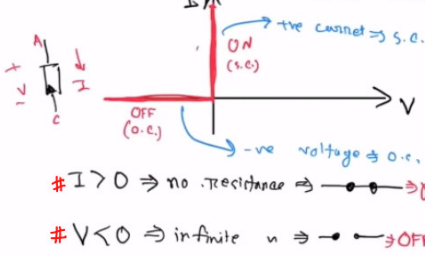
Power generation - AC



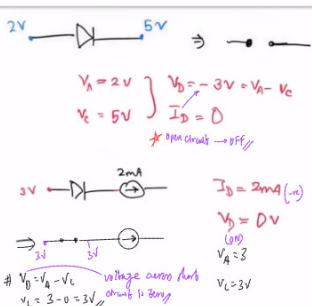
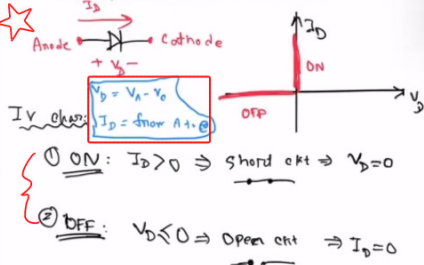
The diagram illustrates the operation of a voltage regulator through several stages:

- Input:** An AC sine wave representing the input voltage V over time t .
- Rectification:** The AC signal is converted into a full-wave rectified signal, shown as a series of positive pulses.
- Filtering:** The rectified signal is filtered by a capacitor, resulting in a signal with "ripple" between the peaks.
- Regulation:** The filtered signal enters a regulation stage, where the output voltage is stabilized at $5V$.
- Feedback:** A feedback loop is shown with a $1V$ reference and a range of $5.5 - 9.5V$.
- Control:** A note at the bottom indicates $0V \rightarrow \text{OFF}$ and $5V \rightarrow \text{ON}$.

What is the JV of an ideal e-valve?



Ideal Diode



Example

5V
2k Ω
I

Equivalent circuit analysis:

Current will try to flow: $\downarrow$ $\rightarrow$ Diode ON

$V = 0V$ $I = \frac{5-0}{2} = 2.5 \text{ mA}$

(c)

5V
2k Ω
I

Diode OFF $\Rightarrow$ $I = 0$

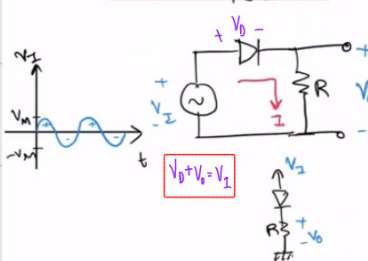
$V_D = -V$

$V_D = V_k - V_A = 0 - (-5) = 5V$

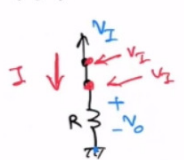
Example 2: $I = 0\text{mA}$
 $V = V_A - V_C = (-5) - 0\text{V} = -5\text{V}$
 current will try to flow: $\uparrow$
 $\therefore$ Diode is OFF
 $I = 0 \rightarrow V = 0$

$$V \neq V_D$$

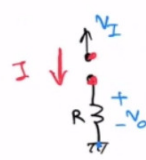
Half Wave Rectifier



the half cycle
ON [conducting]
 $\Rightarrow$ s.c.



-ve half cycle
OFF [not conducting]
 $\Rightarrow$ o.c.



$$V_0 = V_+ - V_- \Rightarrow V_0 = V_2$$

$$I = \frac{V_0}{R} = \frac{V_2}{R}$$

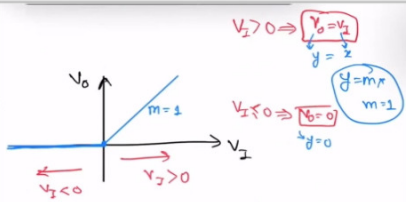
$$V_0 = 0$$

$$I = 0$$

DC to DC transmission is expensive

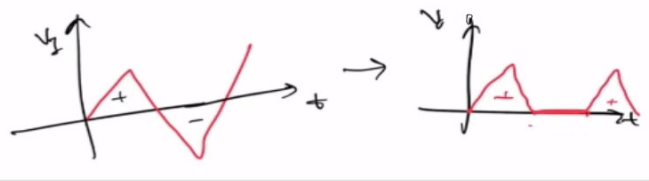
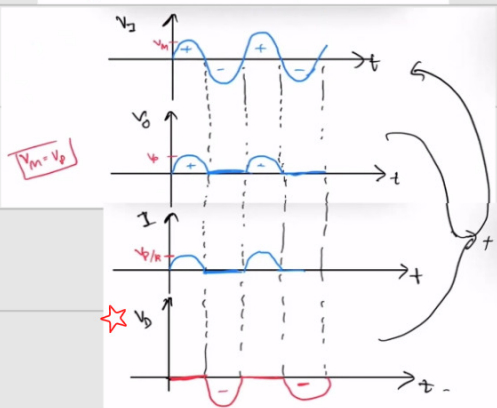
Transfer Characteristics

- ↳ Graph
- ↳ x-axis $\rightarrow$ Input (V/I)
- ↳ y-axis $\rightarrow$ output (V/I)



Waveform

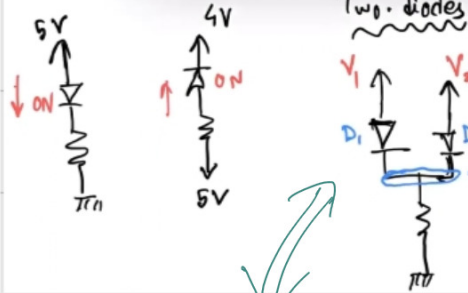
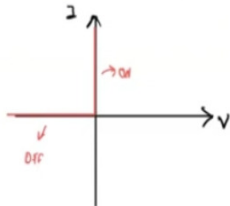
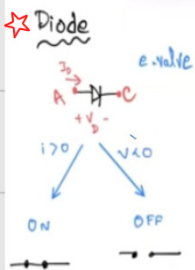
- ↳ Graph
- ↳ x-axis $\Rightarrow$ time
- ↳ y-axis $\Rightarrow V/I$



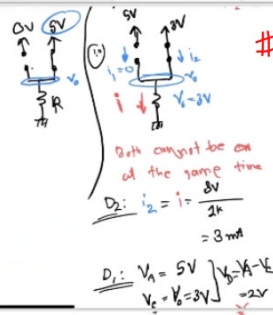
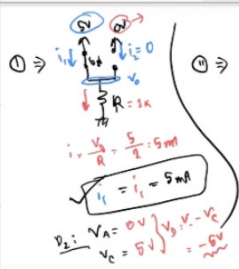
Lecture-6

16/03/2025

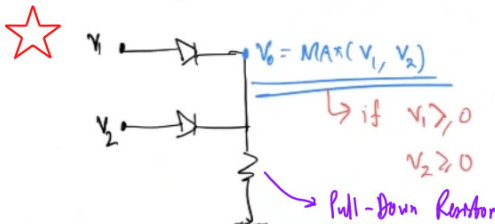
Monday



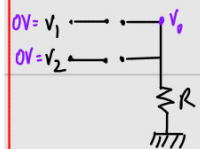
V_1	V_2	D_1	D_2	V_0
① 5V	0V	ON	OFF	$V_0 = 5V$
② 0V	5V	OFF	ON	$V_0 = 5V$
③ 5V	3V	ON	OFF	$V_0 = 5V$
④ 4V	1V	ON	OFF	$V_0 = 4V$
⑤ 3V	10V	OFF	ON	$V_0 = 10V$
⑥ -1V	-3V	OFF	OFF	$V_0 = 0V$



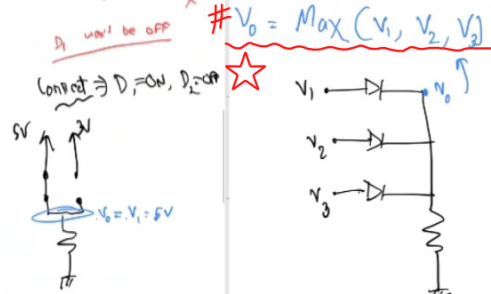
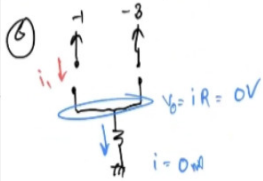
This ckt performs MAX operation betⁿ V_1, V_2



if $V_1, V_2 = 0$

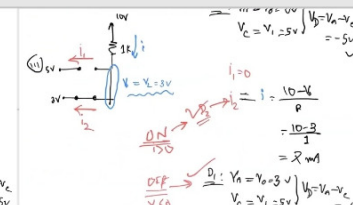
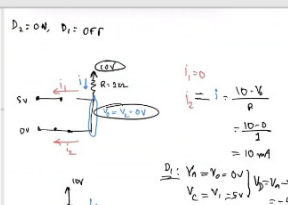
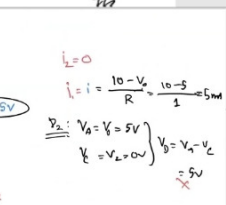
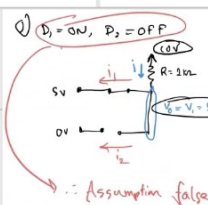


As $V_1, V_2 = 0$, V_0 would be floating node.
 In floating node, voltage is undefined.
 Whenever both diodes are zero, the resistor pulls down the node and makes it zero.

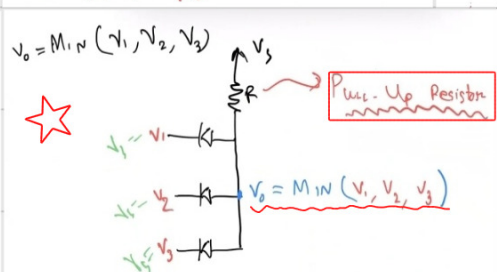
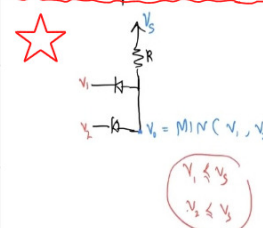


Min

V_1	V_2	D_1	D_2	V_0
① 5V	0V	OFF	ON	$V_0 = 0V$
② 0V	5V	ON	OFF	$V_0 = 0V$
③ 5V	3V	OFF	ON	$V_0 = 3V$
④ 4V	1V	OFF	ON	$V_0 = 1V$
⑤ 3V	10V	ON	OFF	$V_0 = 3V$
⑥ -1V	-3V	OFF	OFF	$V_0 = 0V$



This ckt performs MIN operation betⁿ V_1, V_2



Logic Gates Using Diode

Boolean variable
 $x, y \rightarrow 1, 0$

AND $\rightarrow x \cdot y$
 OR $\rightarrow x + y$
 NOT $\rightarrow \bar{x}, \bar{y}$

How to prepare real boolean variables in CKT?

memorize

① Voltage
 ② Current
 ③ State

most common

0 $\rightarrow$ 0V
 1 $\rightarrow$ 5V

OR

$Z = x + y$
 $V_z = OR(V_x, V_y)$
 $= MAX(V_x, V_y)$

x	y	z = x + y
0	0	0
0	1	1
1	0	1
1	1	1

$V_x = V_y = V_z$

V_x	V_y	V_z
0V	0V	0V
0V	5V	5V
5V	0V	5V
5V	5V	5V

Max

$z = x + y = OR(x, y)$

(Pull down R)

OUT = $x + y + z$

AND

$Z = x \cdot y$
 $V_z = MIN(V_x, V_y)$
 $= MIN(V_x, V_y)$

x	y	z = x \cdot y
0	0	0
0	1	0
1	0	0
1	1	1

$V_x = V_y = V_z$

V_x	V_y	V_z
0V	0V	0V
0V	5V	0V
5V	0V	0V
5V	5V	5V

Min

OUT = $x \cdot y \cdot z$

5V

OUT

Example - 1

Implement using ideal diodes:

OUT = $x + y \cdot z$

Boolean

5V

OUT = $x + y \cdot z$

memorize

resistance can be changed using magnet

Example - 2

Find out as a function of w, x, z

Then simplify.

OUT = $w \cdot x + x + z$

$\star = (w + 1) \cdot x + z$

$= 1 \cdot x + z$

$= x + z$

Example - 2

Find out as a function of w, x, z

Then simplify.

OUT = $w \cdot x + x + z$

$\star = (w + 1) \cdot x + z$

$= 1 \cdot x + z$

$= x + z$

Example - 2

Find out as a function of w, x, z

Then simplify.

OUT = $w \cdot x + x + z$

$\star = (w + 1) \cdot x + z$

$= 1 \cdot x + z$

$= x + z$

Topics

- Review
- Construction of real diode
- Piecewise Linear Models of real diode
- Method of assumed state

Ideal Diode

ON $I_D > 0$
 OFF $V_D < 0$

Rectification

MIN/MAX

$V_z = MAX(V_x, V_y)$

How to make a Diode

① Vacuum tube

② Semiconductor

Group IV: Si, Ge

Group III: P-type

Group V: N-type

Conductivity betⁿ insulation and conductor

$\propto \frac{1}{R}$

Intrinsic (Pure) Si, Ge

Extrinsic (Doped)

17/03/2025

Monday

P-type

majority carrier: hole

n-type

majority carrier: electron

○ hole

● electron

Pn Junction Diode

$F = qE$

Forward Bias

Reverse Bias

The barrier decreases

The barrier increases $I_D \approx 0$

Forward Bias

Reverse Bias

The barrier decreases

The barrier increases $I_D \approx 0$

Real Diode

$I_D = I_S \left[\exp\left(\frac{V_D}{V_T}\right) - 1 \right]$

Reverse saturation current I_S (10⁻⁹ A)

Thermal voltage V_T (0.025 V)

ON: $I_D > 0$
 OFF: $V_D \leq 0$

Piecewise Linear Model of a Real Diode

CVD mode

CVD + P model

ON ($I_D > 0$)
 OFF ($V_D \leq 0$)

We can approximate the I-V characteristics of a real diode with that of an ideal diode.

$r = \frac{1}{\text{slope}}$ steeper slope $\rightarrow$ less R

Example

Find I_D using

- ① Ideal diode model
- ② CVD model with $V_{D0} = 0.2V$
- ③ CVD+P model with $V_{D0} = 0.2V$ and $P = 0.01A$

① $I_D = \frac{5}{2k\Omega} = 2.5mA$

② $V_D = 0.2V$
 $0.2 = V_T - V_D$
 $0.2 = 0.025 - V_D$
 $V_D = 5 - 0.2 = 4.8V$
 $I_D = \frac{4.8}{2k\Omega} = 2.4mA$

③ KVL:

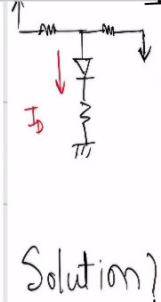
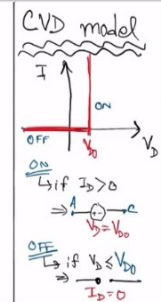
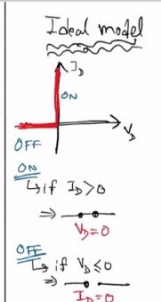
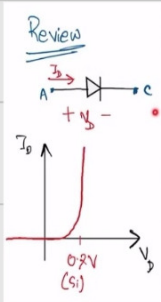
$\sum V_{line} = V_{source} - V_{D0}$

$0.2 + 0.01I_D + 2I_D = 5 - 0$

$I_D(0.01 + 2) = 5 - 0.2$

$I_D = \frac{4.8}{2.01} = 2.39mA$

Lecture-8 (Method of Assumed state)



To find I_D , need to know ON/OFF

To find ON/OFF, need to know I_D

Method of assumed state

Step 1: Assume ON/OFF

Step 2: Solve ON/OFF, I_D ?

Step 3: Verify ON/OFF, I_D ?

If FALSE, go to step 1

Example 1

Find I_1, I_2, I_3, V_1 . Use CVD with $V_{D0} = 1V$

Step 1: ON, ON

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.2 > 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.8 > 0$

Assumption correct!

Example 2

Find I_1, I_2, I_3, V_1 . Use CVD with $V_{D0} = 1V$

Step 1: ON, ON

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.2 > 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.8 > 0$

Assumption wrong!

Step 1: OFF, ON

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow OFF \Rightarrow I_{D1} = 0 < 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.8 > 0$

Assumption correct!

Example-2

Find I_1, I_2, I_3 . Use CVD, $V_{D0} = 1V$

Step 1: ON, OFF

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.2 > 0$
 $D_2 \Rightarrow OFF \Rightarrow I_{D2} = 0 < 0$

Assumption correct!

Example-2

Find I_1, I_2, I_3 . Use CVD, $V_{D0} = 1V$

Step 1: ON, ON

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.2 > 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.8 > 0$

Assumption correct!

Step 1: OFF, ON

Step 2: $V_1 - V_2 = 1V$
 $V_1 - V_2 = 1V$
 $V_1 = 1 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{1}{5} = 0.2mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.2mA$

Step 3: $D_1 \Rightarrow OFF \Rightarrow I_{D1} = 0 < 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.8 > 0$

Assumption correct!

Exercise-1

Find I_1, I_2, V_1 . Use CVD with $V_{D0} = 0.5V$

Step 1: ON, ON

Step 2: $V_1 - V_2 = 0.5V$
 $V_1 - V_2 = 0.5V$
 $V_1 = 0.5 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{0.5}{5} = 0.1mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.1mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.1 > 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.9 > 0$

Assumption correct!

Exercise-2

Find I_1 and I_2 . Use CVD with $V_{D0} = 0.5V$

Step 1: ON, ON

Step 2: $V_1 - V_2 = 0.5V$
 $V_1 - V_2 = 0.5V$
 $V_1 = 0.5 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{0.5}{5} = 0.1mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.1mA$

Step 3: $D_1 \Rightarrow ON \Rightarrow I_{D1} = 0.1 > 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.9 > 0$

Assumption correct!

Step 1: OFF, ON

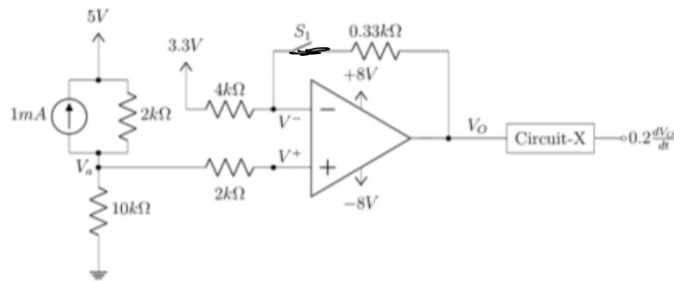
Step 2: $V_1 - V_2 = 0.5V$
 $V_1 - V_2 = 0.5V$
 $V_1 = 0.5 + V_2$
 $I_1 = \frac{V_1 - V_2}{R} = \frac{0.5}{5} = 0.1mA$
 $I_2 = \frac{V_2 - V_3}{R} = \frac{10 - 5}{5} = 1mA$
 $I_3 = \frac{V_3 - V_4}{R} = \frac{5 - 0}{5} = 1mA$
 $I_1 + I_2 = I_3 \Rightarrow I_1 = 0.1mA$

Step 3: $D_1 \Rightarrow OFF \Rightarrow I_{D1} = 0 < 0$
 $D_2 \Rightarrow ON \Rightarrow I_{D2} = 0.9 > 0$

Assumption correct!

Mid Term Practice

The circuit diagram has a switch S_1 which is shown to be 'open' in the figure. The output V_O is passed through an unknown block of 'Circuit-X' and a differentiated result is generated.



- [1 mark] State the equation of gain of an inverting amplifier.
- [3 marks] Calculate the values of V_a and V^+ .
- [2 marks] Determine V_O when the switch S_1 is closed.
- [2 marks] Determine V_O when the switch S_1 is open.
- [2 marks] Design the 'Circuit-X'. Assume any value if necessary.

(a) $(V^+ - V^-)A = V_O \times \text{gain} = -\frac{R_f}{R_i}$ d) $V^+ = 2.5V$

(b) $V_a \left\{ \frac{1}{2} + \frac{1}{10} + \frac{1}{2} \right\} - \frac{5}{2} - \frac{0}{10} - \frac{V^+}{2} + 0 = 0$ $V^- = 3.3V$

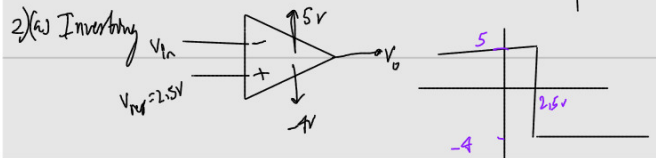
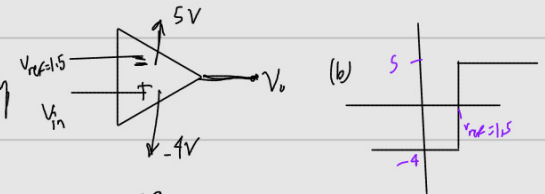
$V_a \left(\frac{3}{5} \right) = \frac{3}{2}$ $\therefore V^- > V^+$

$\therefore V_a < V^+ = 2.5V$ $\therefore V_O = -8V$

(c) $V_O = -\frac{0.33}{4} \times 3.3 = -0.274V$

$V^- \left(\frac{1}{2} + \frac{1}{0.33} \right) - \frac{3.3}{4} - \frac{V_O}{0.33} = 0$

☆ $V^+ = V^- = 2.5V$



Question: 01

An automatic AC switching system using an Op-Amp comparator that will turn on automatically whenever the temperature is higher than 20° Celsius. The output waveform of an Op-Amp comparator should be between the voltage range of $[-4V \ 5V]$. The temperature sensor used by the circuit produces a voltage signal of $1.5V$ for a temperature value of 20° Celsius.

- Design and draw the comparator circuit with the required inputs. [3]
- Draw the VTC curve for your designed comparator. [2]

Question: 02

Design an automatic room heater switching system using an Op-Amp comparator that will turn on automatically whenever the temperature is lower than 20° Celsius. The output waveform of the comparator should be between the voltage range of $[-4V \ 5V]$. The temperature sensor produces a voltage signal of $2.5V$ for a temperature value of 20° Celsius.

- Design and draw the comparator circuit with the required inputs. [3]
- Draw the VTC curve for your designed comparator. [2]

* Linear Amplification only takes place within a valid input range otherwise output will be distorted $\rightarrow$ distortion

The limiting factor of Linear amplification is determined by the power supply to the amplifier.

Features of Ideal Op-Amp

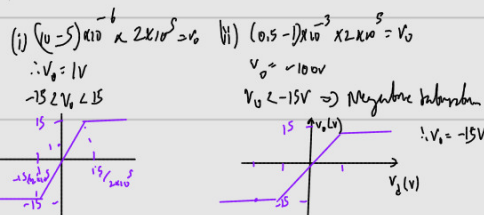
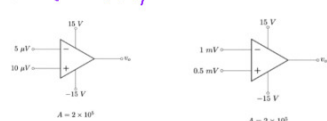
I) Input Resistance ∞ (very high) $\rightarrow$ Input Current 0

II) Output Resistance Very low (0)

III) Amplification factor is very high (∞)

Op-Amp: Examples

Find V_O (Ideal Op-Amp)

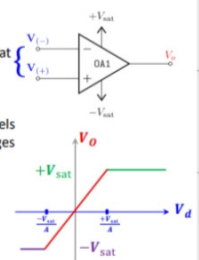


Op-Amp: Summary

Op-Amp **Amplifies** the difference between the voltages at its two input terminals - V_d

However, the **Amplification** is limited within voltage levels defined by the positive and negative saturation voltages $[-V_{sat}, +V_{sat}]$.

The "ideal" op-amp behaves like a **voltage dependent voltage source** within the linear region.



Example - 1

If $v_1 = 0.5V$, calculate:

(a) Output voltage v_o .

(b) Current in the $10k\Omega$ resistor.

(a) $v_o = -\frac{R_f}{R_i} v_1 = -2.5v_1 = -1.25V$

(b) Current through the $10k\Omega$ resistor is $i = \frac{v_1}{R_i} = \frac{0.5}{10} = 50\mu A$

Example - 2

(Inverting Amplifier)

Design the circuit such that the closed loop voltage gain is $A_{CL} = -5$. Assume the op-amp is driven by an ideal sinusoidal source, $v_1 = 0.1 \sin(\omega t)$ (V), that can supply a maximum current of $5\mu A$.

$R_f = \frac{v_o}{i_1} = \frac{0.1}{5 \times 10^{-6}} = 20k\Omega$

$\frac{R_f}{R_i} = 5$

$R_i = 4k\Omega$

Example - 3

Calculate:

(a) Output voltage v_o .

(b) Output current i_o .

(a) $v_o = -\left(\frac{10}{5} + \frac{10}{2.5}\right) = -8V$

(b) $i_o = \frac{v_o}{R_f} = \frac{-8}{10} = -0.8mA$

Example 4

Design an op-amp circuit with inputs v_1 , v_2 and v_3 such that, output voltage v_o is:

Easier Solution:

$v_o = -\frac{2}{5}v_1 - \frac{4}{5}v_2 - \frac{4}{5}v_3$

$= -\frac{1}{5/2}v_1 - \frac{1}{5/4}v_2 - \frac{1}{3/4}v_3$

Difference Amplifier

using superposition principle

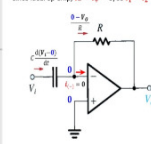
$v_o = v_a = \frac{R_4}{R_3 + R_4} v_2$

$v_o = \left(1 + \frac{R_2}{R_1}\right) v_a - \frac{R_2}{R_1} v_1$

$v_o = \left(1 + \frac{R_2}{R_1}\right) \frac{R_4}{R_3 + R_4} v_2 - \frac{R_2}{R_1} v_1$

Op Amp as Differentiator

Since ideal op-amp, $i_+ = i_- = 0$, so $i_1 = i_2$



Review - Capacitor

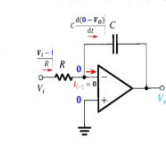
$i_1 = C \frac{dv_1}{dt} = C \frac{dV_i}{dt}$

$\Rightarrow \frac{V_o}{R} = -C \frac{dV_i}{dt}$

$\Rightarrow V_o = -RC \frac{dV_i}{dt}$

Op Amp as Integrator

Since ideal op-amp, $i_+ = i_- = 0$, so $i_1 = i_2$



Review - Capacitor

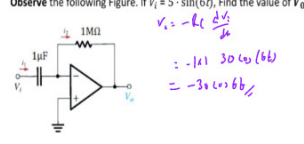
$i_1 = C \frac{dv_1}{dt} = C \frac{dV_i}{dt}$

$\Rightarrow \frac{V_o}{R} = -C \frac{dV_i}{dt}$

$\Rightarrow V_o = -\frac{1}{RC} \int V_i(t) dt$

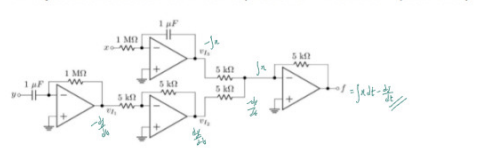
Example 8

Observe the following Figure. If $V_i = 5 \sin(6t)$, Find the value of V_o



Example

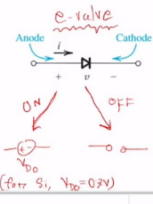
Analyze the circuit below to find an expression of f in terms of inputs x and y .



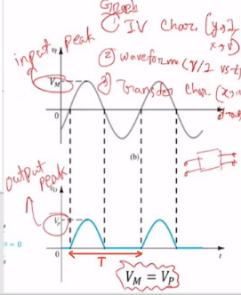
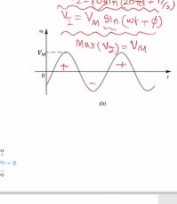
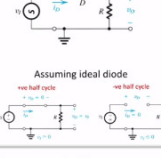
Diode ON: $i > 0, v = V_D$

Diode OFF: $v \leq V_D, i = 0$

Application
1) Rectifier
2) Logic gate

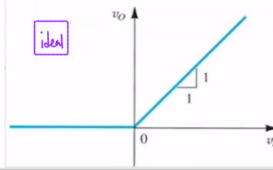


Half wave rectifier

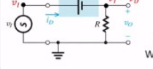


Transfer Characteristics

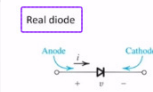
Reminder: Transfer characteristics is a plot (for a circuit) of output vs input
For $v_i > 0, v_o = v_i \Rightarrow y = x$
For $v_i \leq 0, v_o = 0 \Rightarrow y = 0$



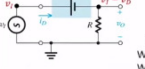
Half wave rectifier



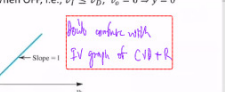
$i_D = \frac{v_i - V_D}{R}$
So diode on when $i_D > 0$ hence when $v_i > V_D$
When ON, i.e., $v_i > V_D, v_o = v_i - V_D \Rightarrow y = x - V_D$
When OFF, i.e., $v_i \leq V_D, v_o = 0 \Rightarrow y = 0$



Half wave rectifier

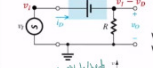


$i_D = \frac{v_i - V_D}{R}$
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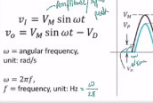


Disadvantage: Unable to utilize the -ve half cycle even though no power loss.

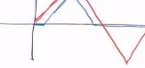
Half wave rectifier



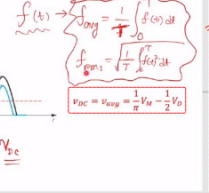
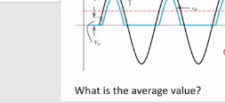
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When OFF, i.e., $v_i \leq V_D, v_o = 0 \Rightarrow y = 0$



Half wave rectifier



$i_D = \frac{v_i - V_D}{R}$
So diode on when $i_D > 0$ hence when $v_i > V_D$
When ON, i.e., $v_i > V_D, v_o = v_i - V_D \Rightarrow y = x - V_D$
When OFF, i.e., $v_i \leq V_D, v_o = 0 \Rightarrow y = 0$



Half wave rectifier

What is the average value? Derivation
$$V_{DC} = V_{avg} = \frac{1}{T} \int_0^T v_o dt$$

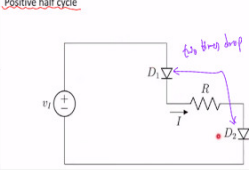
$$= \frac{1}{T} \int_0^{T/2} (V_m \sin \omega t - V_D) dt + \frac{1}{T} \int_{T/2}^T 0 dt$$

$$= \frac{1}{T} \int_0^{T/2} (V_m \sin \omega t) dt - \frac{1}{T} \int_0^{T/2} V_D dt$$

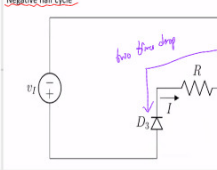
$$= \frac{1}{T} [V_m \times \frac{T}{2}] - \frac{1}{T} [V_D \times \frac{T}{2}]$$

$$= \frac{1}{2} V_m - \frac{1}{2} V_D$$

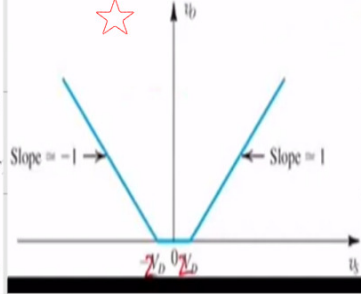
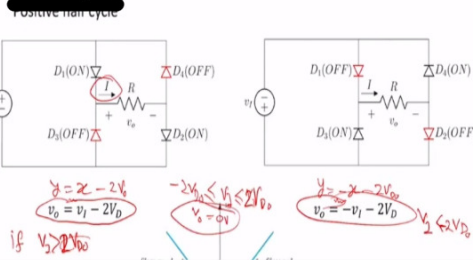
First improvement: Full wave rectifier



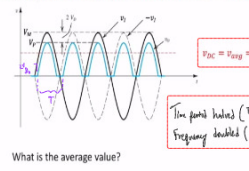
First improvement: Full wave rectifier



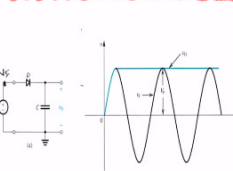
First improvement: Full wave rectifier



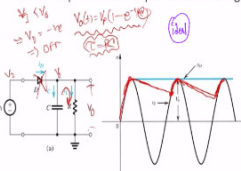
First improvement: Full wave rectifier



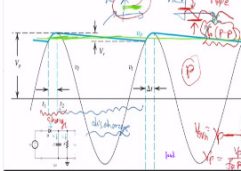
Second improvement: Capacitor smoothing



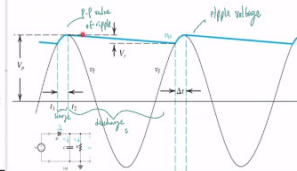
Second improvement: Capacitor smoothing



Second improvement: Capacitor smoothing

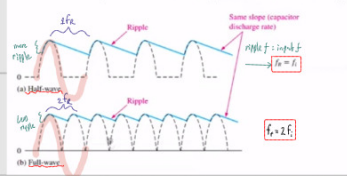


Second improvement: Capacitor smoothing

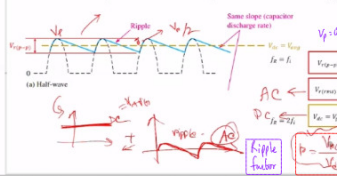


When $V_L < V_{out}$
 $\Rightarrow V_o = -V_L$
Diode OFF
 $\therefore$ capacitor can't charge
 $V_o(t) = V_p(1 - e^{-t/\tau})$
 $\tau = RC$

Second improvement: Capacitor smoothing



Second improvement: Capacitor smoothing



Summary

	Without Capacitor	With Capacitor
Output Peak, V_p	$V_p = V_m$	$V_p = V_m - 2V_{D0}$
Output average, V_{DC}	$V_{DC} = \frac{V_m}{\pi}$	$V_{DC} = \frac{V_m}{\pi} - 2V_{D0}$
Ripple frequency, f_r	$f_r = f$	$f_r = 2f$
Peak to Peak ripple, $V_{r(pp)}$	$V_{r(pp)} = V_m - V_{DC}$	$V_{r(pp)} = V_m - V_{DC}$
RMS of ripple, $V_{r(rms)}$	$V_{r(rms)} = \frac{V_{r(pp)}}{2\sqrt{3}}$	$V_{r(rms)} = \frac{V_{r(pp)}}{2\sqrt{3}}$
Output average, V_{DC}	$V_{DC} = \frac{V_m}{\pi}$	$V_{DC} = \frac{V_m}{\pi} - 2V_{D0}$

Ripple Factor, $r = \frac{V_r(rms)}{V_{DC}}$

Less ripple factor means better rectifier

$I_{DC} = \frac{V_{DC}}{R}$

$V_{rms} = \frac{V_p}{\sqrt{2}}$ (of sine wave not of the ripple)

$V_r(rms) = \frac{V_r(p-p)}{2\sqrt{3}}$ (ripple voltage frequency)

A measure of the fluctuating components is given by the ripple factor r , which is defined as

$r = \frac{\text{rms value of alternating components of the output wave (multimeter in AC mode)}}{\text{average value of the output wave (multimeter in DC mode)}}$

$r = \frac{V_r - rms}{V_{DC}}$

Larger ripple factor means the AC components of the output wave is larger compared to the DC components. Hence, there fluctuation in the output wave is large. Smaller ripple factor means the AC components of the output wave is smaller compared to the DC components. Hence, there fluctuation in the output wave is small. So, the lower the value of the ripple factor, the better the rectifier.

Example - Analysis (Current will be given. Find parameters)

A voltage waveform $v_i = 8\sin(2000\pi t)$ V is input to a full-wave rectifier. A resistance $R = 50k\Omega$ is connected at the load. (Assume that the diodes used in the circuit have a forward drop of 0.8V).

(a) Draw the circuit of the full wave rectifier. Label the input and output voltages properly. [1]

(b) Draw the waveforms of the input and output voltages. What are the peak values of input and output? Show them in the graph. [1+1]

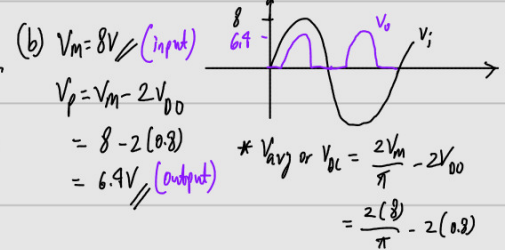
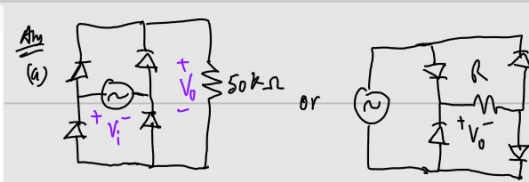
Now a $10\mu F$ capacitor is connected in parallel with the load.

(c) Find the frequency of the input signal and the frequency of the ripple. [0.5+0.5]

(d) Find the peak-to-peak ripple of the output voltage. [1.5]

(e) Find the DC value of the output voltage. [1.5]

(f) If we want to reduce the ripple to 1% of the input peak voltage what value of capacitor should we use? [2]



(c) $\omega = 2000\pi$
 $\omega = 2\pi f$
 $2000\pi = 2\pi f$
 $\therefore f = 1000 \text{ Hz}$

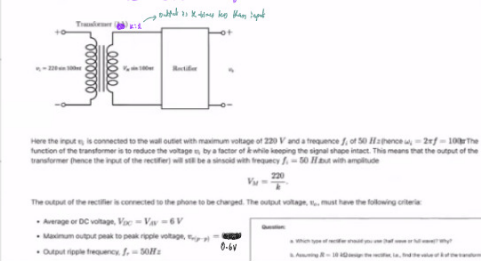
$\therefore f_r = 2 \times f$
 $= 2 \times 1000$
 $= 2000 \text{ Hz}$

(d) $V_{r(pp)} = \frac{V_p}{f_r RC} = \frac{6.4}{2000 \times 50k \times 10^{-6}} = 0.0064 V$

(e) $V_{DC} = V_p - \frac{V_{r(pp)}}{2} = 6.4 - \frac{0.0064}{2} = 6.39 V$

(f) $V_m = 8V$
 $V_{r(pp)} = 8 \times 1\% = 0.08V$
 $\therefore C = 0.8 \mu F$

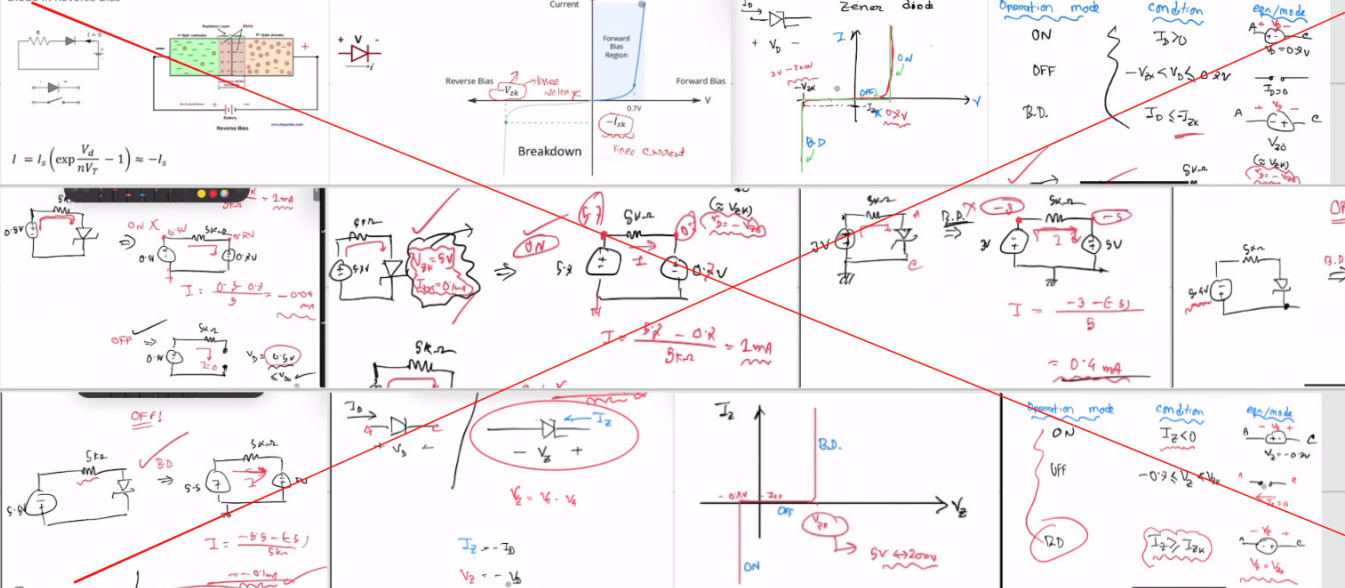
Example - Design (Design a circuit)



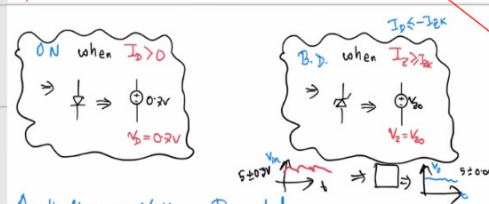
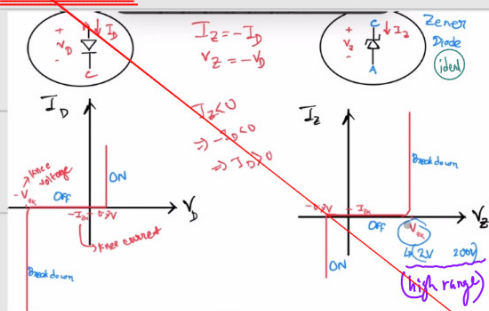
(a) $\omega_i = \omega_r = 50 \text{ Hz}$
 here half wave rectifier

(b) $V_p = V_m - V_{D0}$
 $V_p = 6.3V$
 $V_r = \frac{V_p}{f_r RC}$
 $V_{DC} = 6V$
 $f_r = 50 \text{ Hz}$
 $V_{DC} = V_p - \frac{V_{r(pp)}}{2}$
 $6 = 6.3 - \frac{V_{r(pp)}}{2}$
 $V_{r(pp)} = 0.6V$
 $C = 1 \mu F$ (usually $1 \mu F$)
 $\therefore R = 210k\Omega$

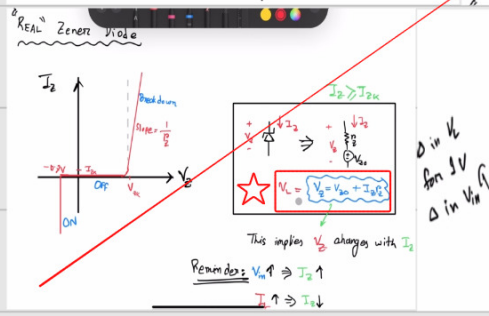
Diode in Reverse Bias



Zener Diode



For regulator to work, Zener **MUST** be in Breakdown
 This means $\Rightarrow V_{in}$ should be high
 $\Rightarrow I_L$ should be low
 From regulator to work properly

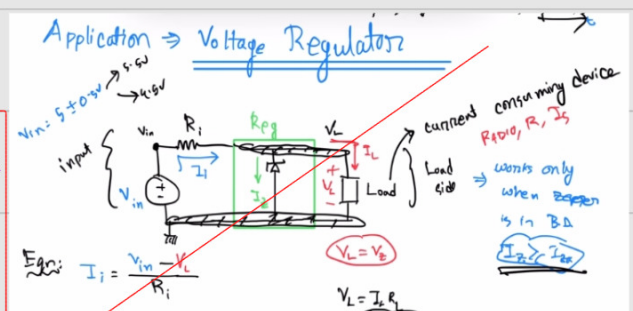
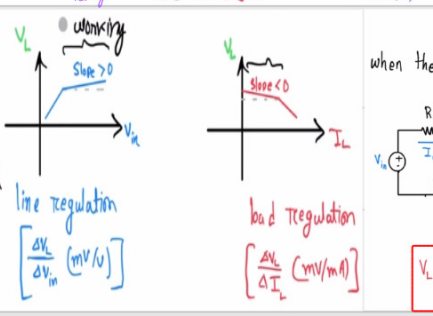


Out of Syllabus

When using diodes, V_D is constant, e.g. 0.7V which might not be enough for the load, hence we need a higher voltage.
 Zener diode behaves like a voltage source in breakdown region but the voltage can be controlled during fabrication.

Design Guideline: Make sure regulator works even in the worst case scenario

- $\Rightarrow V_{in} = V_{in}(MIN)$
- $\Rightarrow I_L = I_L(MIN)$
- $\Rightarrow I_Z = I_Z(MIN) = I_{ZK}$
- $\Rightarrow I_Z \leq I_Z(MIN)$



Application $\Rightarrow$ Voltage Regulator

input $V_{in} = 5 \pm 0.5V \rightarrow 5.5V$

Eqn: $I_i = \frac{V_{in} - V_L}{R_i}$

$I_i = I_Z + I_L$

$\Rightarrow I_Z = I_i - I_L = \frac{V_{in} - V_L}{R_i} - I_L$

$V_L = V_Z$

$V_L \uparrow \Rightarrow I_Z \uparrow$
 $I_Z \downarrow \Rightarrow I_L \uparrow$

What are the values of the slopes?

When the regulator is working

Line regulation = $\frac{\Delta V_L}{\Delta V_{in}} = \frac{R_L}{R_i + R_L}$

Load regulation = $\frac{\Delta V_L}{\Delta I_L} = -\frac{R_L}{R_i + R_L}$

performance measure

L-1.1 | Part-2

Example

The output of rectifier has a DC value of 10 V and a fluctuation of ± 1 V. This voltage is fed as the input V_{in} into the voltage regulator shown below:

$10 \pm 1V = 9V$ to $11V$

The Zener diode is specified as $r_z = 50 \Omega$ and $V_{Z0} = 8V$. The knee current of the Zener is 0.5 mA.

- What can be the minimum and maximum values of the input voltage V_{in} ? [1]
- In worst case condition, find the Zener current I_Z and the Zener voltage V_Z . [1+1]
- What is the maximum value of R_i that we can use if the Zener has to stay in breakdown even in the worst case condition? [3]
 Hint: What is the voltage and current of R_i in the worst case conditions?
- Calculate the value of line regulation and load regulation for this regulator. [3]
- If the load current increases by 0.5 mA, by how much will the load voltage decrease? [1]

Example 4.7

The 6.8-V Zener diode in the circuit of Fig. 4.21(a) is specified to have $V_Z = 6.8V$ at $I_Z = 5mA$, $r_z = 20 \Omega$, and $I_{ZK} = 0.2mA$. The supply voltage V^+ is nominally 10 V but can vary by $\pm 1V$.

- Find V_o with no load and with V^+ at its nominal value.
- Find the change in V_o resulting from the $\pm 1V$ change in V^+ . Note that $(\Delta V_o / \Delta V^+)$, usually expressed in mV/V, is known as **line regulation**.
- Find the change in V_o resulting from connecting a load resistance R_L that draws a current $I_L = 1mA$, and hence find the **load regulation** $(\Delta V_o / \Delta I_L)$ in mV/mA.
- Find the change in V_o when $R_L = 2k\Omega$.
- Find the value of V_o when $R_L = 0.5k\Omega$.
- What is the minimum value of R_i for which the diode still operates in the breakdown region?

$\Rightarrow V_Z = 6.8 = V_{Z0} + I_Z R_Z$

$\Rightarrow 6.8 = V_{Z0} + 5 \times 0.02$

$\Rightarrow V_{Z0} = 6.7V$

$R_Z = 0.02k\Omega$

$I_Z = \frac{10 - 6.8}{0.5 + 0.02} = 6.35mA$

$V_Z = V_{Z0} + I_Z R_Z = 6.8V$

$\frac{R_L}{R_i + R_L} = -\frac{R_L}{R_i + R_L}$

① $\min \{V_{in}\} = 10 - 1 = 9V$

$\max \{V_{in}\} = 10 + 1 = 11V$

② Worst case

① $V_{in} = \min(V_{in}) = 9V$

② $I_L = \max(I_L) = 9.5mA$

③ $I_Z = I_{ZK} = 0.5mA$

$R_i = \frac{V_{in} - V_Z}{I_Z + I_L}$

$R_i = \frac{9V - 8V}{0.5mA + 9.5mA}$

$R_i = \frac{1V}{10mA} = 0.1k\Omega$

④ line regulation = $\frac{\Delta V_L}{\Delta V_{in}} = \frac{R_L}{R_i + R_L}$

$\frac{\Delta V_L}{\Delta V_{in}} = \frac{0.05}{0.05 + 0.1} = 0.333 V/V$

$= 333 mV/V$

load regulation = $\frac{\Delta V_L}{\Delta I_L} = -\frac{R_L}{R_i + R_L}$

$\frac{\Delta V_L}{\Delta I_L} = -\frac{0.05 \times 0.1}{0.05 + 0.1} = -0.0833 V/mA$

$= -83.3 mV/mA$

$\frac{\Delta V_L}{\Delta I_L} = -33.3 mV/mA$

$\Delta I_L = 0.5mA$

$\Rightarrow \Delta V_L = -33.3 \times 0.5 = -16.65 mV$

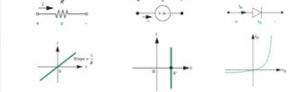
Zener must be in breakdown for regulator to work

Lecture -13

05/05/2025
Nancy

Three terminal devices

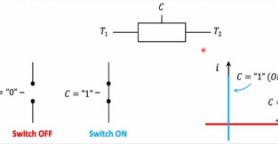
- Two terminal devices have fixed IV characteristics



- Three terminal devices - IV of two terminal can be controlled using the third terminal.
- Examples - Switch (linear), Transistors (non-linear)

Switch - IV Characteristics

- IV characteristics between terminal T_1 and T_2 is controlled by C



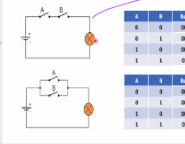
Switch - Types

- Depending on the control, the switch can be:
 - Analog: Controlled using physical toggle/button
 - Digital: Controlled using voltage or current. Example - MOSFET (voltage controlled), BIT (current controlled)

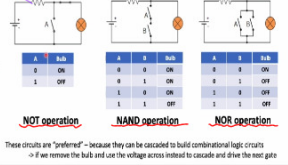


Switch Application - Logic Gates

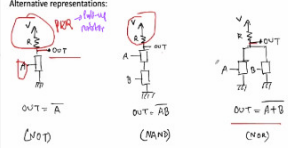
- We can use switches to build logic gates



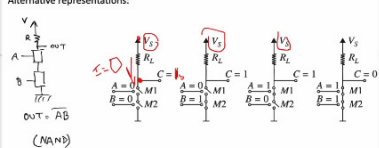
Switch Application - Logic Gates



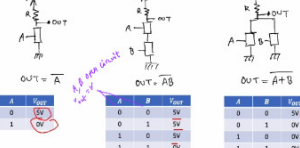
Switch Application - Logic Gates



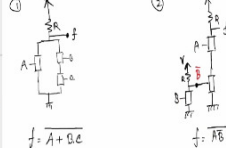
Switch Application - Logic Gates



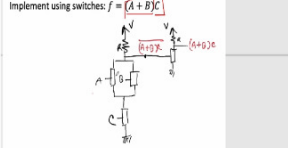
Switch Application - Logic Gates



Examples

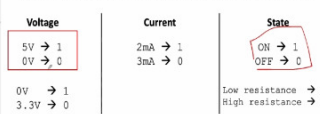


Example

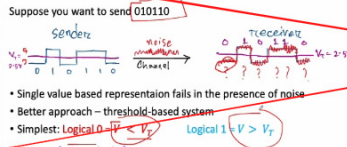


Digital Representation

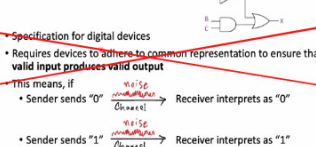
- Binary -> 2 states (0/False, 1/True)
- Binary variables in circuit, need to use two states of device/parameters



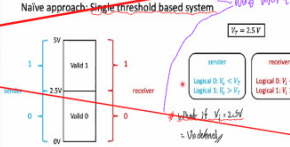
Digital Representation



Static Discipline

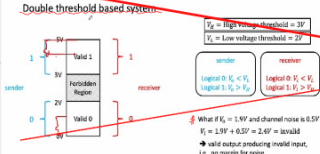


Static Discipline

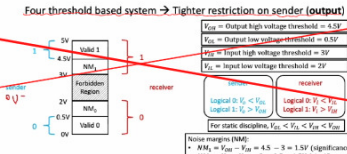


Stable Discipline
Out of Syllabus

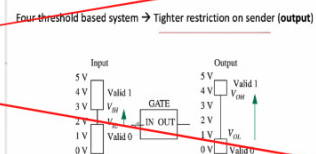
Static Discipline



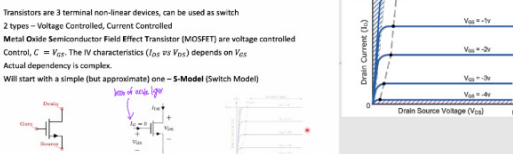
Static Discipline



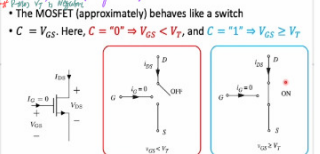
Static Discipline



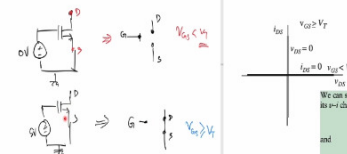
Transistors as Digital Switch



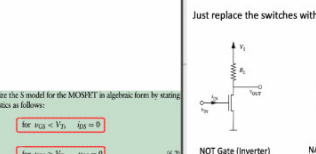
MOSFET S-Model



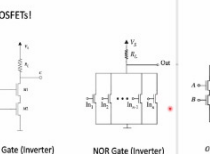
MOSFET S-Model



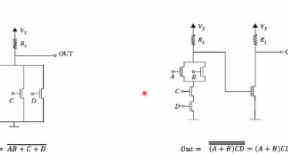
MOSFET S-Model



Logic Gates using MOSFET



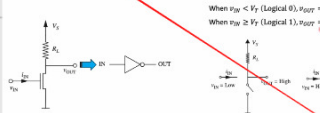
MOSFET Logic Gates - More Examples



Lecture -14

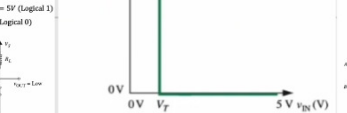
Voltage Transfer Characteristics (VTC)

- Reminder: VTC is a graph where x axis = input voltage, y axis = output voltage
- Why? Design logic gates to follow a given static discipline



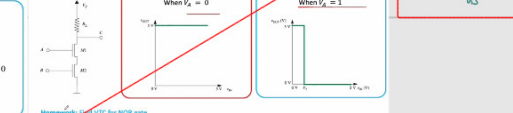
VTC of NAND gate

- We only have one x axis, but two inputs
- Solution: Draw two VTC, one considering $V_{A1} = 0$, one considering $V_{A1} = 1$

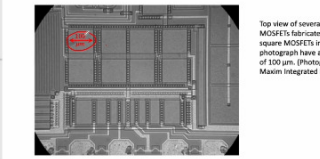


VTC of NAND gate

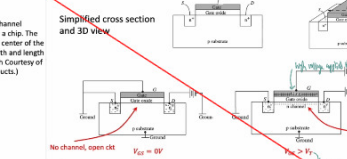
- We only have one x axis, but two inputs
- Solution: Draw two VTC, one considering $V_{A1} = 0$, one considering $V_{A1} = 1$



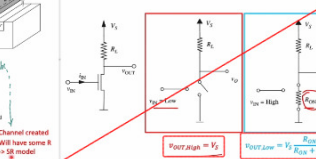
Construction of Real MOSFET



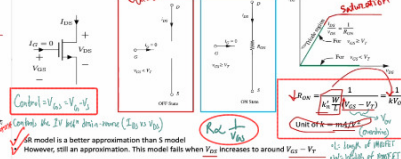
Construction of Real MOSFET



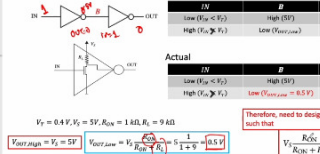
SR Model - Inverter



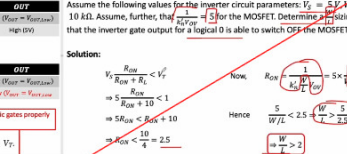
SR Model



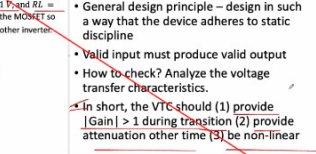
Design of logic gates



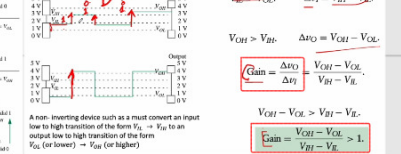
Design of logic gates - Example



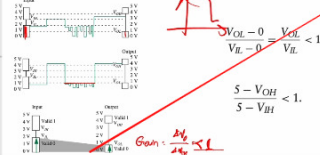
Static Analysis



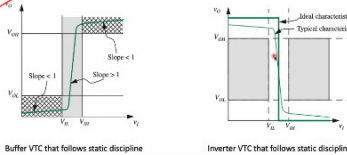
Static Analysis and VTC - Gain



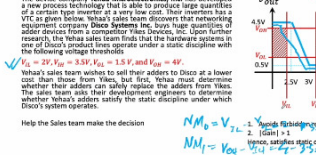
Static Analysis and VTC - Attenuation



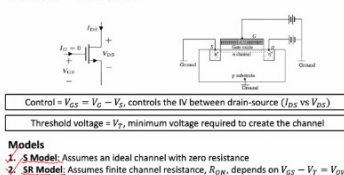
Static Discipline and VTC



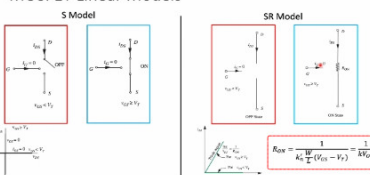
Static Discipline and VTC - Example



Review - MOSFET

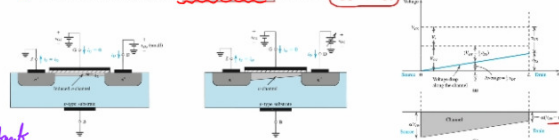


MOSFET Linear Models

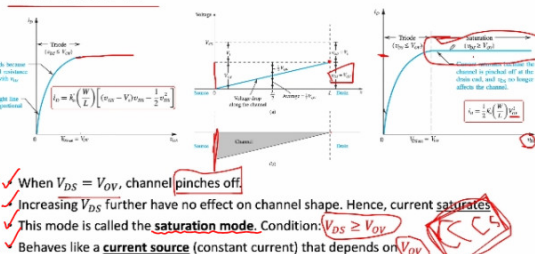


Real MOSFET

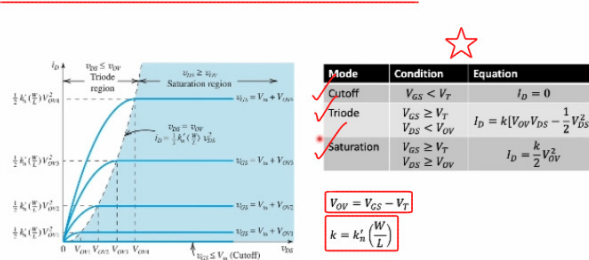
- Why $R_{ON} = \frac{1}{k_n \frac{W}{L} (V_{GS} - V_T)} = \frac{1}{kV_{OV}}$? Because channel width $\propto V_{OV}$, and $R \propto \frac{1}{\text{width}}$
- For small V_{DS} , uniform channel, hence fixed R_{ON} , therefore SR model valid.
- As V_{DS} is increased, channel becomes tapered cause $V_{GD} \downarrow$. Resistance $\uparrow$, slope $\downarrow$.
- This mode is called the **triode mode**. Condition: $V_{DS} < V_{OV}$



Real MOSFET



IV Characteristics of Real MOSFET



Solving Circuits with MOSFET

- Use Method of Assumed State!
- Three steps:
 - Assume: One of the modes (Cutoff, Triode, Saturation)
 - Solve: Use corresponding equation and KCL+KVL
 - Verify: Check if the conditions of V_{GS} and V_{DS} are satisfied. If not, repeat.
- Might need to solve quadratic equation ($ax^2 + bx + c = 0$).
- If we get two roots, choose the one that's **favorable** to your assumption.

Example 1

Solution:

Step 1: Assume the MOSFET is in saturation.

Step 2: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$. Here, $V_{GS} = V_G - V_S = 0 - 0 = 0$, $V_T = 1$.
 Therefore, $I_D = \frac{k_n}{2} (0 - 1)^2 = \frac{k_n}{2}$.
 Apply $V_{DS} = V_D - V_S = 2 - 0 = 2$.
 Check: $V_{DS} = 2 > V_{OV} = 1$. Therefore, assumption correct.
 Correct ans: $I_D = 0.25 \text{ mA}$, $V_D = 2 \text{ V}$.

Example 2

Solution:

Step 1: Assume the MOSFET is in saturation.

Step 2: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$. Here, $V_{GS} = V_G - V_S = 5 - 0 = 5$, $V_T = 1$.
 Therefore, $I_D = \frac{k_n}{2} (5 - 1)^2 = 8k_n$.
 Apply $V_{DS} = V_D - V_S = 5 - 0 = 5$.
 Check: $V_{DS} = 5 > V_{OV} = 4$. Therefore, assumption correct.
 Correct ans: $I_D = 8k_n$, $V_D = 5 \text{ V}$.

Example 2

Solution:

Step 1: Assume the MOSFET is in saturation.

Step 2: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$. Here, $V_{GS} = V_G - V_S = 5 - 0 = 5$, $V_T = 1$.
 Therefore, $I_D = \frac{k_n}{2} (5 - 1)^2 = 8k_n$.
 Apply $V_{DS} = V_D - V_S = 5 - 0 = 5$.
 Check: $V_{DS} = 5 > V_{OV} = 4$. Therefore, assumption correct.
 Correct ans: $I_D = 8k_n$, $V_D = 5 \text{ V}$.

Example 3

Solution:

Step 1: Assume the MOSFET is in saturation.

Step 2: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2$. Here, $V_{GS} = V_G - V_S = 5 - 0 = 5$, $V_T = 1$.
 Therefore, $I_D = \frac{k_n}{2} (5 - 1)^2 = 8k_n$.
 Apply $V_{DS} = V_D - V_S = 5 - 0 = 5$.
 Check: $V_{DS} = 5 > V_{OV} = 4$. Therefore, assumption correct.
 Correct ans: $I_D = 8k_n$, $V_D = 5 \text{ V}$.

Practice

Question 4 [CO1, CO4]

Analyze the following circuit to find the value of I_D and V_{DS} using the Method of Assumed State. You must validate your assumption.

Hint: Use the I_D vs V_{DS} characteristic. Use Ohm's law to represent V_{GS} and V_{DS} in terms of x .

Solution:

Assume $I_D = x$. For Sat: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2 = 10 - 5x$.
 For Cutoff: $I_D = 0$.
 For Sat: $I_D = \frac{k_n}{2} (V_{GS} - V_T)^2 = 10 - 5x$.
 Therefore, $I_D = 10 - 5x$.
 Solve for x , take the one that's **favorable** to your assumption.

When gate connected to drain $\rightarrow$ saturation mode



Tip! If $V_G \approx V_D$ then saturation mode

$V_{GS} < V_{Th} \rightarrow$ Subthreshold

Current controlled logic gate not used as bad variable

(No SR Model in BJT)

Bipolar Junction Transistor

Current-controlled transistor, 3 terminals - Base, Emitter, Collector

- IV between C and E (I_C vs V_{CE}) is controlled by base current, I_B
- IV quite similar to MOSFET, but there are some differences
- We can use a S-model here too, but controlled by I_B (instead of V_{GS})

BJT vs MOSFET - Differences

BJT

- Current controlled, I_B controls I_C vs V_{CE}
- Base current, I_B , is the control. Hence $I_C \approx \beta I_B$, rather than $I_C \approx I_B$
- Current in active region changes linearly with control I_B . Hence, $I_C \approx \beta I_B$
- Slope in the saturation region is high, hence very small resistance when "ON"

MOSFET

- Voltage controlled, V_{GS} controls I_D vs V_{DS}
- Gate current, I_G , is always zero. Therefore, $I_D \approx I_{GS}$
- Current in saturation region changes quadratically with control V_{GS} . Hence, $I_D \approx \frac{k_n}{2} (V_{GS} - V_T)^2$
- Slope in the saturation region is low, hence significant resistance when "ON"

BJT vs MOSFET - Similarities

BJT

- Saturation mode for small V_{CE} ($< 0.2V$)
- Open circuit in Cutoff mode ($I_B = 0$)
- Can use as a switch - S-Model!

MOSFET

- Triode mode for small V_{DS} ($< V_{OV}$)
- Approximately short circuit in Triode mode ($V_{DS} \approx 0$)
- Open circuit in Cutoff mode ($V_{GS} < V_T$, or $V_{GS} = V_T$)
- Can use as a switch - S-Model!

BJT S-Model

The BJT (approximately) behaves like a switch (V_{CE} small).

$C = I_B$. Here, $C = "0" \Rightarrow I_B = 0$, and $C = "1" \Rightarrow I_B \approx I_{B,sat}$

Logic 0: $I_B = 0 \Rightarrow I_C = 0$

Logic 1: $I_B = I_{B,sat} \Rightarrow I_C \approx I_{C,sat}$

Current-Controlled Logic Gates using BJT

Current-controlled logic gate is not used as bad variable

Major problem: Cannot cascade (Why?)

For I_B to be voltage source, need to use V_{BE} as I_B

BJT Inverter (NOT Gate)

BJT NAND Gate

BJT NOR Gate

From Current Controlled to Voltage Controlled

Controlled by carrying a resistor

$V = IR$

$V_{BE} = I_B R$

$V_{CE} = I_C R$

$V_{DS} = I_D R$

From Current Controlled to Voltage Controlled

Controlled

$V_{BE} = I_B R$

$V_{CE} = I_C R$

$V_{DS} = I_D R$

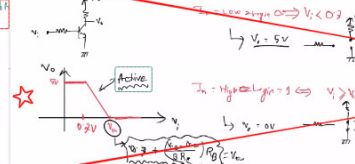
Voltage Controlled Logic Gates

Can't be too high or too low

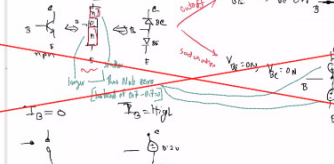
NAND

NOR

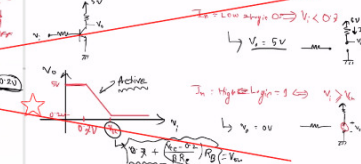
Voltage Transfer Characteristics - S-model



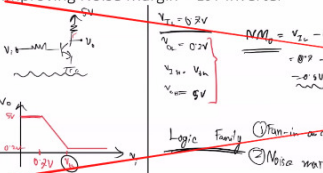
BJT SV Model



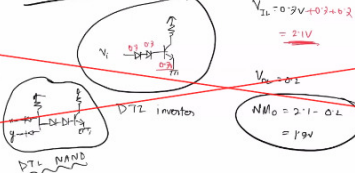
Voltage Transfer Characteristics - SV-model



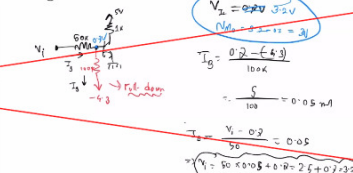
Improving Noise Margin - BJT Inverter



DTL Logic



RTL Logic



BJT acts as a switch in saturation mode (ON) and cutoff region (OFF). Active region is used for amplification.

L-17 Part-01

Physical Structure

Physical Structure: NPN (Emitter, Base, Collector) and PNP (Collector, Base, Emitter).
Circuit Symbol: NPN (arrow out), PNP (arrow in).
Trade representation: NPN (arrow out), PNP (arrow in).

Simplified operation of BJT

BE → ON, OFF; BC → ON, OFF. 4 combinations, 4 modes of operation.

Mode	BE	BC
Cutoff	OFF	OFF
Saturation	ON	ON
Active	ON	OFF
Reverse active	OFF	ON

Case 1: BE → OFF and BC → OFF

$I_B = 0$, $I_E = 0$, $I_C = 0$. Cutoff.

Case 2: BE → ON and BC → OFF

Active Mode. $I_C = \beta I_B$, $I_E = I_C + I_B$. $I_E = I_C(1 + \frac{1}{\beta})$.

Case 3: BE → ON and BC → ON

Saturation mode. $I_C < \beta I_B$. $I_E = I_C + I_B$.

When BJT is in Active Mode

I_C behaves like a current controlled current source. $I_C = \beta I_B$. $I_E = I_C + I_B$.

L-17 Part-02

Bipolar Junction Transistors (BJT)

- Three terminal devices
- Used: Amplifiers, Switch, Logic Gates
- Two types: NPN, PNP

Modes of operation

Mode	BE	BC
Active	ON	OFF
Saturation	ON	ON
Cutoff	OFF	OFF
Reverse active	OFF	ON

Condition and Model

State: Active, Saturation, Cutoff. Conditions: $I_B > 0$, $V_{BE} > 0.7V$, $V_{CE} > 0.2V$.

Method of assumed state

- Assume: ACTIVE [$V_{BE} = 0.7V$, $I_C = \beta I_B$]
- Solve: [KCL, KVL, nodal]
- Verify: [ACTIVE $V_{CE} > 0.2V$, SAT $I_C / I_B < \beta$, CUT $V_{BE} < 0.7V$ and $V_{CE} < 0.2V$]

Example-1

Find V_B for $V_E = 0V$, $V_C = 10V$, $\beta = 100$.
Assume Active: $V_{BE} = 0.7V$, $I_C = \beta I_B$.
Assume Sat: $I_C < \beta I_B$.
Assume Cut: $V_{BE} < 0.7V$ and $V_{CE} < 0.2V$.

L-17 Part-03

Example-1

Find V_B for $V_E = 0V$, $V_C = 10V$, $\beta = 100$.
Assume Active: $V_{BE} = 0.7V$, $I_C = \beta I_B$.
Assume Sat: $I_C < \beta I_B$.
Assume Cut: $V_{BE} < 0.7V$ and $V_{CE} < 0.2V$.

Example-2

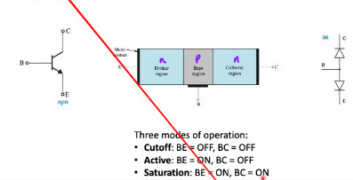
Find V_B for $V_E = 0V$, $V_C = 10V$, $\beta = 100$.
Assume Active: $V_{BE} = 0.7V$, $I_C = \beta I_B$.
Assume Sat: $I_C < \beta I_B$.
Assume Cut: $V_{BE} < 0.7V$ and $V_{CE} < 0.2V$.

Example-3

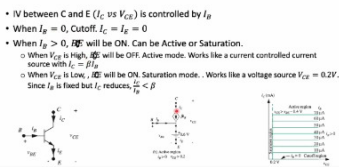
Find V_B for $V_E = 0V$, $V_C = 10V$, $\beta = 100$.
Assume Active: $V_{BE} = 0.7V$, $I_C = \beta I_B$.
Assume Sat: $I_C < \beta I_B$.
Assume Cut: $V_{BE} < 0.7V$ and $V_{CE} < 0.2V$.

Lecture - 18

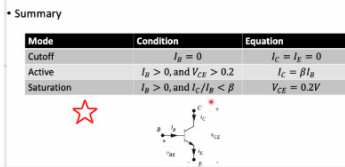
BJT Review



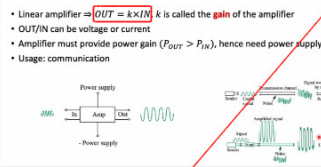
BJT Review



BJT Review

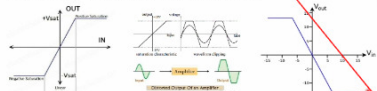


Amplifier



Transfer Characteristics of Amplifiers

- $y = mx$, hence straight line going through origin. Here, $m = k = \text{gain}$
- Saturation due to limited power supply. Output cannot be greater than P.S.
- Input must be within a **valid input range**, otherwise output will be distorted
- For inverting amplifiers, gain is negative. Hence $y = -mx$



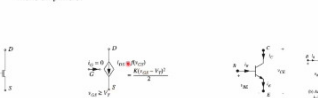
Types of Amplifier and Circuit Realization

- Depending on input (V or I) and output (V or I) = 4 types
- 1. Voltage amplifier (in = V, Out = V)
- 2. Current amplifier (in = I, Out = I)
- 3. Transconductance amplifier (in = V, Out = I)
- 4. Transresistance amplifier (in = I, Out = V)

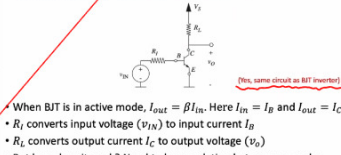


Making Dependent Sources

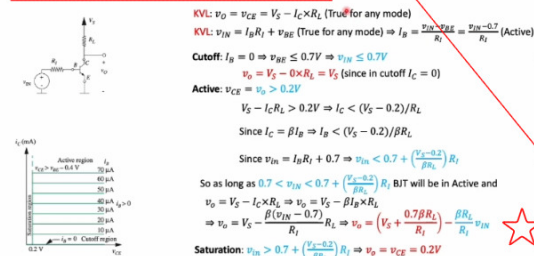
- MOSFET in Saturation mode acts like a voltage controlled current source
- BJT in Active mode acts like a current controlled current source
- Since amplifiers are made using dependent sources, we can use transistors to make amplifiers!



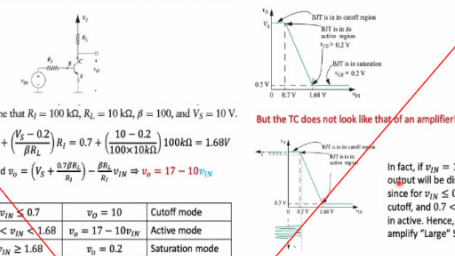
BJT Common Emitter Amplifier



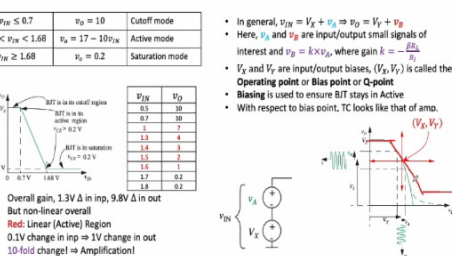
Finding VTC of CE Amplifier



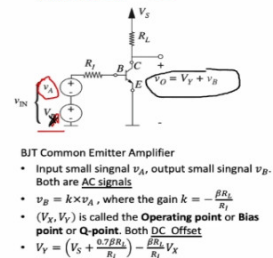
VTC of CE Amplifier



Small Signal Amplification

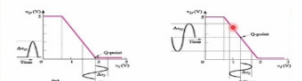


Summary



Valid voltage range The range of input for which the BJT in the circuit operate in the Active region.

If not maintained, output might be distorted



Biased, but **Valid voltage range** not maintained

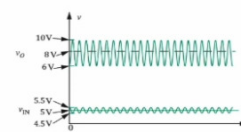
- Need to select Q-point or Bias point in such a way that input is within **Valid voltage range**
- In this case, **Valid voltage range**: $0.7 < v_{IN} < 0.7 + \left(\frac{V_S - 0.7}{\beta R_L}\right) R_L$
- How to select Q-point? Best option: middle of active region, which will give maximum swing (peak-to-peak) for v_o

Example 1

- Consider the BJT common emitter circuit with $\beta = 100$, $R_1 = 100k\Omega$, $R_L = 10k\Omega$ and $V_S = 10V$
- We want to set bias the point in such a way that the DC portion of the input, i.e., $V_X = 1.2V$.
- Under small signal approximation, if the input $v_{IN} = V_X + v_i(t)$, the output will be $v_o = V_Y + k v_i(t)$.
- This means the output will be some DC value V_Y plus the amplified version of the small signal $v_i(t)$.
- Here assume that $v_i = 0.1 \cos \omega t$.
- What is the value of DC part of the output, i.e., V_Y ?
- $V_X = 1.2$, therefore $V_Y = \left(V_S + \frac{0.7\beta R_L}{R_1}\right) - \frac{\beta R_L}{R_1} V_X = 5$
- What will be the value of gain k under small signal approximation?
- $k = -\frac{\beta R_L}{R_1} = -10$
- What is amplitude (peak-to-peak) of the input small signal waveform $v_i(t)$?
- $\max = 0.1V, \min = -0.1V$. Therefore, **peak-to-peak amplitude = 0.2V**
- What is amplitude (peak-to-peak) of the output small signal waveform?
- Amplitude of output = $|k| \times \text{Amplitude of input} \Rightarrow 10 \times 0.2 = 2V$**

Example 2

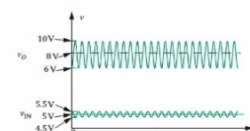
The input voltage of a common source amplifier is given as $v_{IN} = V_X + v_i(t)$ and the output voltage is given $v_o = V_Y + k v_i(t)$. Here, $v_i(t)$ is a sinusoidal voltage with amplitude a , V_X is the input DC offset voltage, V_Y is the output DC offset voltage. The input and output waveforms are given below. Notice the output small signal is inverted compared to input small signal. Hence, the small signal gain, k , will be negative.



- What is the amplitude of the input small signal $v_i(t)$?
- 0.5V**
- What is the amplitude of the output small signal?
- 2V**
- Hence, what is the small signal gain k ?
- $k = -\frac{2}{0.5} = -4$
- From the above graph, what is the value of input DC voltage V_X and the output DC voltage V_Y ?
- $V_X = 5V, V_Y = 8$

Example 2

The input voltage of a common source amplifier is given as $v_{IN} = V_X + v_i(t)$ and the output voltage is given $v_o = V_Y + k v_i(t)$. Here, $v_i(t)$ is a sinusoidal voltage with amplitude a , V_X is the input DC offset voltage, V_Y is the output DC offset voltage. The input and output waveforms are given below. Notice the output small signal is inverted compared to input small signal. Hence, the small signal gain, k , will be negative.

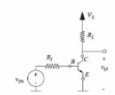


- Design the circuit, i.e., find the value of V_S , R_1 , and R_L to achieve given input-output voltage relation. Given $\beta = 100$.

$$k = -4 = -\frac{\beta R_L}{R_1} \Rightarrow \frac{\beta R_L}{R_1} = 4 \Rightarrow R_L = 0.04 R_1$$
$$\text{Let } R_1 = 100k\Omega \Rightarrow R_L = 4k\Omega$$
$$V_Y = \left(V_S + \frac{0.7\beta R_L}{R_1}\right) - \frac{\beta R_L}{R_1} V_X \Rightarrow 8 = \left(V_S + \frac{0.7 \times 100 \times 10^3 \times 4 \times 10^3}{100 \times 10^3}\right) - \frac{100 \times 10^3 \times 4 \times 10^3}{100 \times 10^3} \times 5$$
$$\Rightarrow V_S = 8 + 20 - 2.8 = 25.2V$$

Example 3

Choose an operating point for the amplifier below to maximize the input voltage swing. Here $R_1 = 100k\Omega$, $R_L = 10k\Omega$, $\beta = 100$, $V_S = 10V$



Solution: BJT will be active for the **valid input range**: $0.7 < v_{IN} < 0.7 + \left(\frac{V_S - 0.7}{\beta R_L}\right) R_L$

$$\text{Here, } 0.7 + \left(\frac{V_S - 0.7}{\beta R_L}\right) R_L = 0.7 + \left(\frac{10 - 0.7}{100 \times 10}\right) \times 10 = 0.7 + 0.98 = 1.68V$$

Therefore, **valid input range**: $0.7V < v_{IN} < 1.68V$

For maximum input swing, V_X should be midway between the **valid input range**

$$\text{Hence, } V_X = \frac{0.7 + 1.68}{2} = 1.19V$$

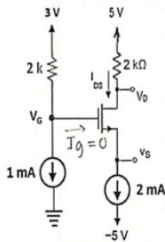
$$\text{Therefore, } V_Y = \left(V_S + \frac{0.7\beta R_L}{R_1}\right) - \frac{\beta R_L}{R_1} V_X = \left(10 + \frac{0.7 \times 100 \times 10}{100}\right) - \frac{100 \times 10}{100} \times 1.19 = 5.1$$

Therefore, maximum peak-to-peak input swing = $(1.68 - 0.7) = (1.68 - 1.19) = 0.58V$

Note: $V_Y = 5.1$ is midway between the **valid output range**: $0.2 \leq v_o < 10$

Question 01 (CO3) points)

Refer to the MOSFET in the figure below where $V_T = 1\text{ V}$ and $K = k_n' W/L = 4\text{ mA/V}^2$.

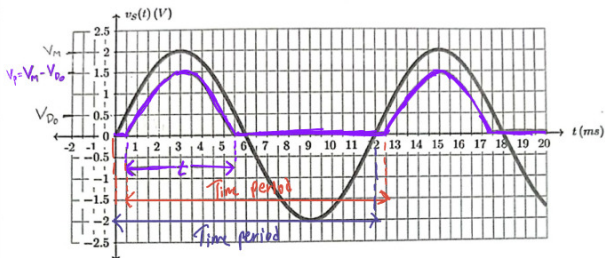


- [CO1] Calculate the value of the gate voltage V_G and the drain-source current I_{DS} .
- [CO2] Calculate the drain voltage V_D .
- [CO2] Analyze the circuit to find the source voltage V_S . [Use the method of assumed states.]

1-071

Question 01 Out of 02:

$v_S(t)$ is the input voltage to a half wave rectifier without filter capacitor.



- [4 marks] Assuming a cut-in voltage of 0.5 V for the diode used in the rectifier, approximately draw the output voltage waveform on the same grid.
- [2 marks] Determine the approximate average of the output voltage, V_{avg} or V_{DC} .
- [2 marks] Determine the fraction of time within a cycle the diode is conducting. [Determine from the above graph]

Bonus: Determine the average or DC voltage if a capacitor of $1\text{ }\mu\text{F}$ is connected in parallel with the load resistor. [2], $R_L = 10\text{ k}\Omega$

Question 02 Out of 02:

- Draw the IV curve of a MOSFET considering the S-model only. [2]
- Design a circuit using ideal MOSFETs (S-model) to implement the logic function: [3]

$$(1) F = (A + \overline{B}) \cdot (\overline{B} + B \cdot C)$$

Practice

$$(a) V_G\left(\frac{1}{2}\right) - \frac{3}{2} + 1 = 0$$

$$V_G = 1\text{ V}$$

$$I_{D1} = 2\text{ mA}$$

$$(b) \frac{5 - V_D}{2} = I_{D1}$$

$$\therefore V_D = 1\text{ V}$$

(c) Assume saturated

$$V_{GS} > V_T \quad I_{D1} = \frac{1}{2} K V_{GS}^2, \quad V_{DS} = V_{GS} - V_T$$

$$V_{GS} > V_{DS}$$

$$2 = \frac{1}{2} \times 4 \times (-V_D)^2$$

$$V_D = 1, -1$$

$$\# \text{ If } V_D = 1\text{ V} \quad \# \text{ If } V_D = -1\text{ V}$$

$$V_{GS} = 1 - 1 = 0 < V_T \quad V_{DS} = 1 - (-1) = 2$$

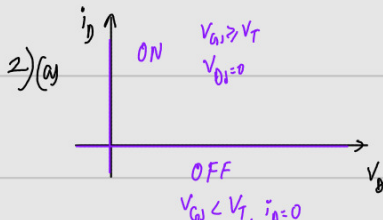
$$\therefore \text{ rejected} \quad \therefore V_{GS} > V_T$$

$$\therefore V_D = -1\text{ V}$$

$$(1) (a) V_P = V_m - V_{D0}$$

$$= 2 - 0.5$$

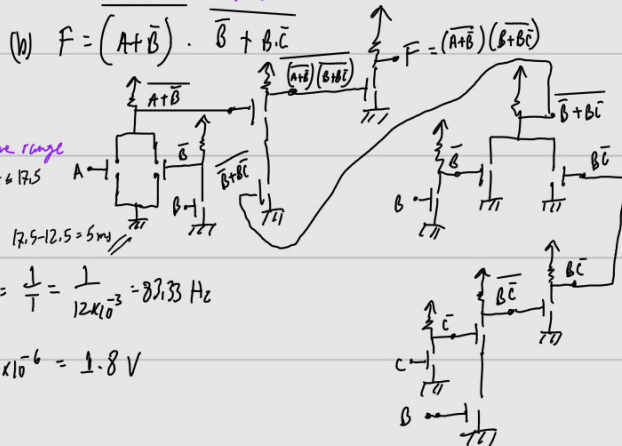
$$= 1.5\text{ V}$$



$$(b) V_{DC} = \frac{V_m}{\pi} - \frac{V_{D0}}{2}$$

$$= \frac{2}{\pi} - \frac{0.5}{2}$$

$$= 0.387\text{ V}$$



$$(c) 0.5 \leq t \leq 5.5 \text{ ms and } 12.5 \leq t \leq 17.5 \text{ ms}$$

$$\therefore \text{Conducting time} = 5.5 - 0.5 = 5 \text{ ms or } 17.5 - 12.5 = 5 \text{ ms}$$

$$\text{Bonus: } V_{DC} = V_P - \frac{V_r(P-P)}{2}, \quad f_r = \frac{1}{T} = \frac{1}{12 \times 10^{-3}} = 83.33 \text{ Hz}$$

$$V_r(P-P) = \frac{V_P}{f_r \times R \times C} = \frac{1.5}{83.33 \times 10^3 \times 1 \times 10^{-6}} = 1.8\text{ V}$$

$$\therefore V_{DC} = 1.5 - \frac{1.8}{2} = 0.6\text{ V}$$

Problem 6 [From PDS YT Video]

- The output voltage of a full wave rectifier has been filtered with two $2.5\text{ }\mu\text{F}$ capacitors in parallel. The peak, average, and frequency of the output voltage are 4.3 V , 4 V , and 300 Hz respectively. Determine the load resistance.

$$C = 2.5\text{ }\mu\text{F} \quad V_{avg} = V_P - \frac{V_r(P-P)}{2} \quad \# C_{eq} = 2.5 + 2.5 = 5\text{ }\mu\text{F}$$

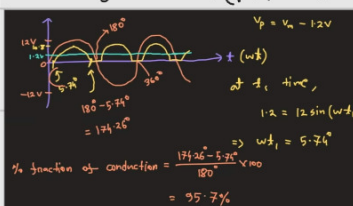
$$V_P = 4.3\text{ V}$$

$$V_{avg} = 4\text{ V} \quad 4 = 4.3 - \frac{V_r(P-P)}{2} \quad V_r(P-P) = \frac{V_P}{f_r R C} = 0.16 = \frac{4.3}{300 \times R \times 5 \times 10^{-6}} \quad \therefore R = 48\text{ k}\Omega$$

$$f = 300\text{ Hz} \quad \therefore V_r(P-P) = 0.16\text{ V}$$

Problem 8

- A 24 V peak to peak sine wave ac voltage is to be rectified with a full wave bridge rectifier circuit without filter capacitor. Determine the fraction of time each diode is conducting when on and the average output voltage. Assume all the diodes are identical with a cut-in voltage $V_{D0} = 0.6\text{ V}$.



$$V_{avg} = \frac{2V_m}{\pi} - 2V_{D0}$$

$$= \frac{2 \times 12}{\pi} - 2(0.6)$$

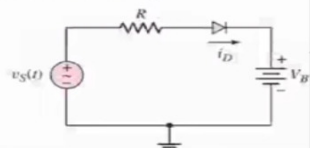
$$= 6.4\text{ V}$$

Problem 10

- A battery V_B is charged through a half wave rectifier circuit as shown below. The voltage of the battery at the time of charging is 4.5 V . The resistance R is $250\text{ }\Omega$. $v_S(t)$ is a 24 V peak to peak sine wave. Cut-in voltage for the diode is $V_{D0} = 0.6\text{ V}$.

- What is the required value of $v_S(t)$ for the diode to conduct?
- Determine the peak diode current, maximum reverse-bias diode voltage, and the fraction of the cycle over which the diode is conducting.

$$iii) V_{avg} = V_{DC} = ?$$

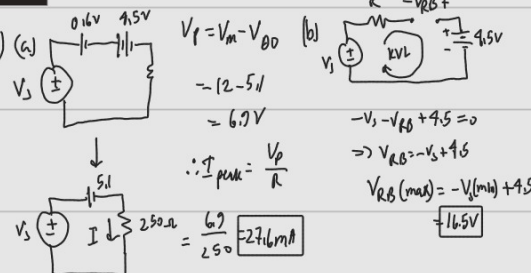


$$(i) \text{ Diode will be on if } i_D > 0 \quad (ii) (a) \quad V_S - V_{D0} - V_B > 0$$

$$\Rightarrow V_S > V_{D0} + V_B$$

$$\Rightarrow V_S > 0.6 + 4.5$$

$$\therefore V_S > 5.1\text{ V}$$



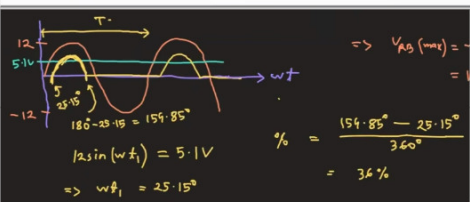
$$V_P = V_m - V_{D0}$$

$$= 12 - 0.6$$

$$= 11.4\text{ V}$$

$$\therefore I_{peak} = \frac{V_P}{R}$$

$$= \frac{11.4}{250} = 45.6\text{ mA}$$



$$iii) V_{avg} = V_{DC} = ?$$

$$= \frac{1}{2\pi} \int_0^{2\pi} (12 \sin \omega t - 0.6) d\omega t$$

$$= \frac{1}{2\pi} \left(-12 \cos \omega t + 0.6 \omega t \right) \Big|_0^{2\pi}$$

$$= \frac{1}{2\pi} \left(-12 \cos 2\pi + 0.6 \cdot 2\pi - (-12 \cos 0 + 0.6 \cdot 0) \right)$$

$$= \frac{1}{2\pi} \left(-12 + 1.2\pi - (-12) \right)$$

$$= \frac{1.2\pi}{2\pi} = 0.6\text{ V}$$

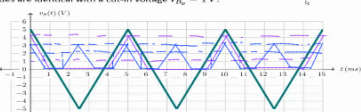
Problem 11

- The following is the input voltage to a half wave rectifier without filter capacitor. Draw the output waveform on the same grid. Assume a cut-in voltage $V_{D0} = 1.0\text{ V}$ for the diode.



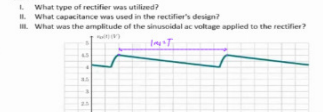
Problem 14

- Draw the output voltage waveforms if the following is the input voltage to a (i) half wave rectifier and (ii) full wave rectifier, both without filter capacitor. Assume all the diodes are identical with a cut-in voltage $V_{D0} = 1\text{ V}$.



Problem 13

- The following is the rectified output of an ac sinusoidal voltage at 1 kHz frequency, provided to a load of $4.5\text{ k}\Omega$. The diode(s) used in the rectifier had a cut-in voltage of 0.5 V .



$$(i) f = \frac{1}{T} = 1\text{ kHz}$$

$$\therefore \text{same frequency as input, so half wave}$$

$$(ii) V_P = \frac{V_r}{FRC}$$

$$0.5 = \frac{4.5}{1000 \times 4.5 \times 10^3}$$

$$\therefore V_{avg} = \frac{V_P}{2} = 2.25\text{ V}$$

Problem 1

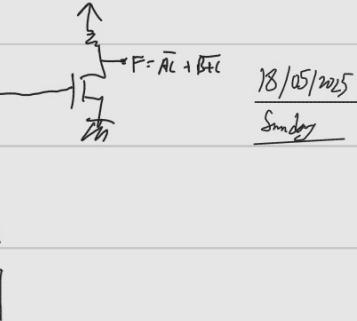
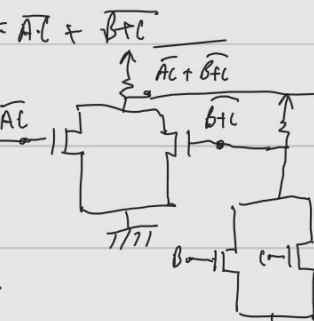
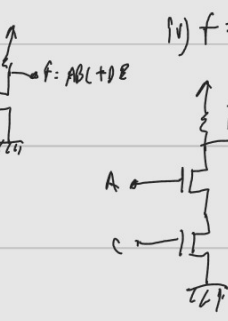
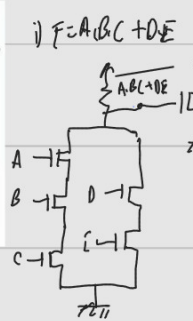
- Give a switch-MOSFET implementation of the following logic functions. A, B, C, D, E, and F are Boolean inputs.
- I. $f = A \cdot B \cdot C + D \cdot E$
- II. $f = \overline{A} \cdot B \cdot (\overline{C} + D)$
- III. $f = A \cdot B + \overline{A} \cdot \overline{B}$
- IV. $f = \overline{A} \cdot \overline{C} + \overline{B} + \overline{C}$
- V. $f = (A \cdot B + C) \cdot D$
- VI. $f = A \cdot B + C \cdot D$
- VII. $f = A \cdot B \cdot C + D$
- VIII. $f = (A + B) \cdot (C + D)$
- IX. $f = (A \cdot B + C) \cdot D \cdot (E + F)$
- X. $f = A \oplus B$
- XI. $f = \overline{C} \cdot (A + B) \cdot (A + \overline{B} + C)$

$$A + B = \overline{\overline{A} \cdot \overline{B}}$$

$$C \cdot C = C$$

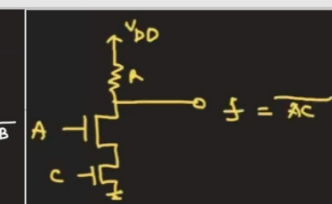
$$\overline{A \cdot B} = \overline{A} + \overline{B}$$

De Morgan's Law

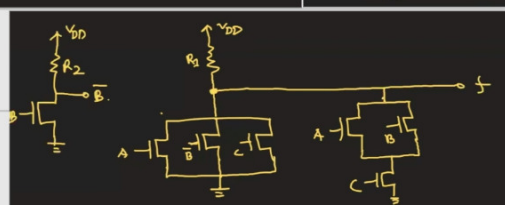


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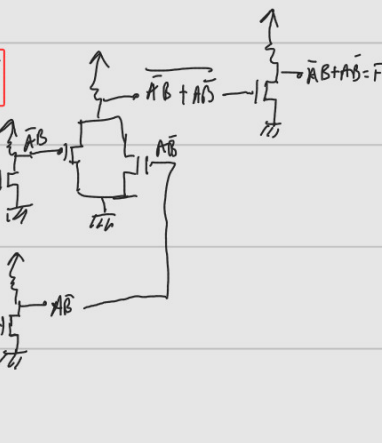
$$\begin{aligned} f &= \overline{A \cdot C + B \cdot C} = \overline{A \cdot C} \cdot \overline{B \cdot C} = \overline{A \cdot C} \cdot \overline{B \cdot C} \\ &= \overline{A \cdot C} \cdot (\overline{B} + \overline{C}) = \overline{A \cdot C} \cdot \overline{B} + \overline{A \cdot C} \cdot \overline{C} \\ &= \overline{A \cdot C} \cdot \overline{B} + \overline{A \cdot C} \cdot \overline{C} = \overline{A \cdot C} \cdot (\overline{B} + \overline{C}) = \overline{A \cdot C} \end{aligned}$$



$$\begin{aligned} (XI) \quad f &= \overline{C \cdot (A + B)} \cdot (A + B + C) \\ &= \overline{C \cdot (A + B)} \cdot (A + B + C) \\ &= \overline{C \cdot (A + B)} + (A + B + C) \end{aligned}$$

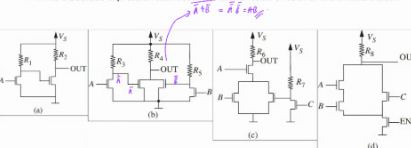


$$X) \quad f = A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$$



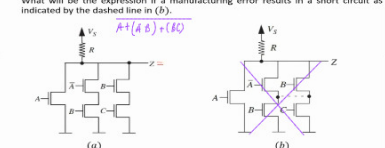
Problem 2

- Write a Boolean expression that describes the function of each of the circuits below.



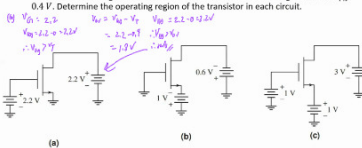
Problem 9

- Write a Boolean expression that describes the function of the circuit in figure (a). What will be the expression if a manufacturing error results in a short circuit as indicated by the dashed line in (b).



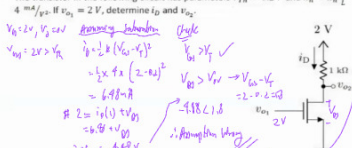
Problem 10

- The threshold voltage for each of the transistors in the following circuits is $V_{th} = 0.4 \text{ V}$. Determine the operating region of the transistor in each circuit.



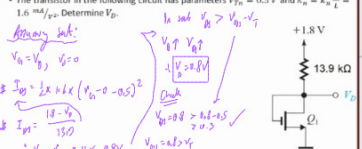
Problem 12

- The transistor in the following circuit has parameters $V_{th} = 0.2 \text{ V}$ and $k_n = k_p = 200 \mu\text{A/V}^2$. Determine I_D and V_{DS} .



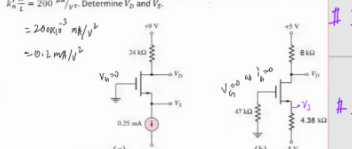
Problem 13

- The transistor in the following circuit has parameters $V_{th} = 0.5 \text{ V}$ and $k_n = k_p = 1.6 \text{ mA/V}^2$. Determine V_D .



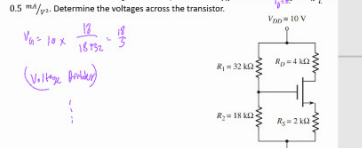
Problem 15

- The transistors in the following circuit has parameters $V_{th} = 0.6 \text{ V}$ and $k_n = k_p = 200 \mu\text{A/V}^2$. Determine V_D and V_S .



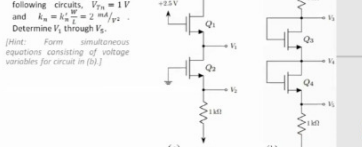
Problem 18

- The transistor in the following circuit has parameters $V_{th} = 0.8 \text{ V}$ and $k_n = k_p = 0.5 \text{ mA/V}^2$. Determine the voltages across the transistor.



Problem 23

- For the transistors in the following circuits, $V_{th} = 1 \text{ V}$ and $k_n = k_p = 2 \text{ mA/V}^2$. Determine V_1 through V_5 .



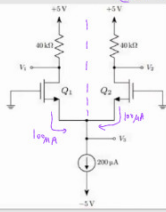
$$\begin{aligned} (b) \quad V_0 &= V_0, V_{th} > V_0 \rightarrow \text{sat} \\ I_{D1} &= \frac{10 - V_0}{1} \quad (1) \\ I_{D2} &= \frac{V_0 - 0}{1} = V_0 \quad (2) \\ (3) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (3) \\ (4) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (4) \\ (5) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (5) \\ (6) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (6) \\ (7) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (7) \\ (8) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (8) \\ (9) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (9) \\ (10) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (10) \\ (11) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (11) \\ (12) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (12) \\ (13) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (13) \\ (14) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (14) \\ (15) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (15) \\ (16) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (16) \\ (17) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (17) \\ (18) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (18) \\ (19) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (19) \\ (20) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (20) \\ (21) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (21) \\ (22) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (22) \\ (23) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (23) \\ (24) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (24) \\ (25) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (25) \\ (26) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (26) \\ (27) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (27) \\ (28) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (28) \\ (29) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (29) \\ (30) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (30) \\ (31) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (31) \\ (32) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (32) \\ (33) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (33) \\ (34) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (34) \\ (35) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (35) \\ (36) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (36) \\ (37) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (37) \\ (38) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (38) \\ (39) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (39) \\ (40) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (40) \\ (41) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (41) \\ (42) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (42) \\ (43) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (43) \\ (44) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (44) \\ (45) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (45) \\ (46) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (46) \\ (47) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (47) \\ (48) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (48) \\ (49) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (49) \\ (50) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (50) \\ (51) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (51) \\ (52) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (52) \\ (53) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (53) \\ (54) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (54) \\ (55) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (55) \\ (56) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (56) \\ (57) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (57) \\ (58) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (58) \\ (59) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (59) \\ (60) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (60) \\ (61) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (61) \\ (62) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (62) \\ (63) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (63) \\ (64) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (64) \\ (65) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (65) \\ (66) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (66) \\ (67) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (67) \\ (68) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (68) \\ (69) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (69) \\ (70) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (70) \\ (71) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (71) \\ (72) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (72) \\ (73) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (73) \\ (74) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (74) \\ (75) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (75) \\ (76) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (76) \\ (77) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (77) \\ (78) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (78) \\ (79) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (79) \\ (80) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (80) \\ (81) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (81) \\ (82) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (82) \\ (83) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (83) \\ (84) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (84) \\ (85) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (85) \\ (86) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (86) \\ (87) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (87) \\ (88) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (88) \\ (89) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (89) \\ (90) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (90) \\ (91) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (91) \\ (92) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (92) \\ (93) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (93) \\ (94) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (94) \\ (95) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (95) \\ (96) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (96) \\ (97) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (97) \\ (98) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (98) \\ (99) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (99) \\ (100) \quad I_{D1} &= \frac{1}{2} \times 2 \times (V_0 - 1)^2 \quad (100) \end{aligned}$$

$$\begin{aligned} V_3 &= 6 \text{ V}, V_5 = 4 \text{ V}, V_4 = 2 \text{ V} \\ \text{Condition must be } V_3 > V_4 > V_5 \\ \text{but } 6 > 2 > 4 \end{aligned}$$

$$\begin{aligned} (8) \quad V_{D1} & > V_{th} - V_t \\ 7.55 - 5 & > 1 - 0.4 \\ 2.55 & > 0.6 \end{aligned}$$

Problem 24

For the transistors in the following circuits, $V_{BE} = 0.7V$, $\beta = 100$, $V_{BE}(\text{Active}) = 0.7V$, $V_{BE}(\text{Sat}) = 0.8V$, $V_{CE}(\text{Sat}) = 0.2V$. Determine V_E through V_{CE} .



Assuming Sat (both)

$$100 = \frac{I_C}{I_B} \Rightarrow I_C = 100 I_B$$

$$\Rightarrow (V_E - 0.7) = 1$$

$$V_E = 1.7V$$

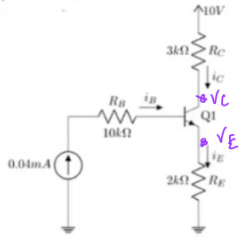
$$V_E = 1.7V$$

$$V_E = 1.7V$$

$$V_E = 1.7V$$

BJT Problems

Analyze the following circuit to find the values of I_B , I_C , I_E and V_{CE} .



$\beta = 100$
 $\alpha = 0.99$
 $V_{BE}(\text{Active}) = 0.7V$
 $V_{BE}(\text{Sat}) = 0.8V$
 $V_{CE}(\text{Sat}) = 0.2V$

Assuming Active

$$V_{BE} = 0.7V, I_C = \beta I_B, I_E = \frac{I_C}{\alpha}$$

$$I_B = 0.04A, I_C = \alpha I_E$$

$$I_C = \beta I_B = 100 \times 0.04 = 4A$$

$$\therefore I_E = 4 + 0.04 = 4.04A$$

$$I_C = \frac{10 - V_C}{3}$$

$$V_C = -2V$$

$$\frac{V_E - 0}{2} = 4.04$$

$$\therefore V_E = 8.08V$$

$$V_{CE} = V_C - V_E$$

$$= -2 - 8.08$$

$$= -10.08V$$

$$V_{CE} > V_{CE}(\text{Sat})$$

$$-10.08 > 0.2$$

$$\therefore \text{Assumption wrong}$$

Assuming Sat

$$V_{BE} = 0.8V, \frac{I_C}{I_B} < \beta$$

$$V_{CE} = 0.2V$$

$$I_B = 0.04A$$

$$\# 10 - 0 = I_C(3) + V_{CE} + I_E(2)$$

$$10 = 3I_C + 0.2 + 2(I_C + 0.04)$$

$$\therefore I_C = 1.94mA$$

$$\therefore \frac{I_C}{I_B} = 48.6 < \beta \therefore \text{Active mode Assumption valid}$$

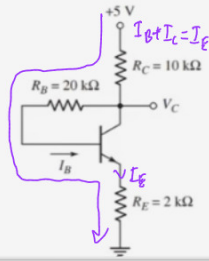
$$\# \frac{V_E - 0}{2} = I_E \# \frac{5 - V_C}{10} = I_B + I_C$$

$$V_E = 0.702V, V_C = 1.99V$$

$$\therefore V_{CE} = 1.99 - 0.702 = 1.288 > 0.2$$

$$\therefore \text{Active mode}$$

Determine the values of V_C and I_B .



$\beta = 75$
 $V_{BE}(\text{Active}) = 0.7V$
 $V_{BE}(\text{Sat}) = 0.8V$
 $V_{CE}(\text{Sat}) = 0.2V$

Assuming Active

$$V_{BE} = 0.7V, V_{CE} > 0.2V, I_E = \frac{I_C}{\alpha} = (\beta + 1)I_B$$

$$\beta = 75$$

$$\# 5 - 0 = 10(I_B + I_C) + 20I_B + 0.7 + 2I_E = 0$$

$$5 - 0.7 = 10(I_B + \beta I_B) + 20I_B + 2(\beta + 1)I_B$$

$$\therefore I_B = 4.61\mu A$$

$$I_C = 75 \times 4.61 \times 10^{-6}$$

$$= 0.346mA$$

$$I_E = I_C + I_B = 0.351mA$$

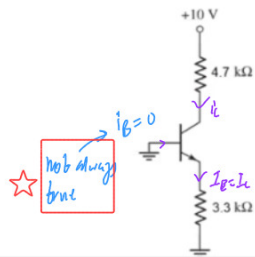
$$\# \frac{V_E - 0}{2} = I_E \# \frac{5 - V_C}{10} = I_B + I_C$$

$$V_E = 0.702V, V_C = 1.99V$$

$$\therefore V_{CE} = 1.99 - 0.702 = 1.288 > 0.2$$

$$\therefore \text{Active mode}$$

Determine the collector to emitter voltage.



$\beta = 75$
 $V_{BE}(\text{Active}) = 0.7V$
 $V_{BE}(\text{Sat}) = 0.8V$
 $V_{CE}(\text{Sat}) = 0.2V$

Assume Active mode

$$V_{BE} = 0.7V$$

$$V_E = 0 - 0.7 = -0.7V$$

$$I_E = \frac{-0.7 - 0}{3.3} = -0.21mA$$

$$\therefore I_E \text{ can't be -ve as current flows from collector to emitter Active mode}$$

$$\therefore \text{Not active}$$

Assuming Sat

$$V_{BE} = 0.8V, V_{CE} = 0.2V$$

$$V_E = -0.18V$$

$$I_E = \frac{-0.18}{3.3}$$

$$\therefore I_E < 0$$

$$\therefore \text{Also Not Sat}$$

Cut off Check

$$V_{BE} \leq 0.7V, V_{CE} \leq 0.5V$$

$$I_C = 0, I_E = 0, I_B = 0$$

$$V_C = 10V, V_B = 0$$

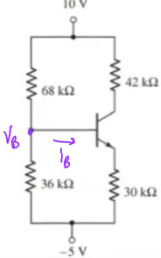
$$V_E = 0V$$

$$\therefore V_{BE} < 0.7V \checkmark$$

$$V_{BC} = 0 - 10 = -10 < 0.5V \checkmark$$

$$\therefore \text{Cut-off}$$

Determine the collector to emitter voltage.



$\beta = 50$
 $V_{BE}(\text{Active}) = 0.7V$
 $V_{BE}(\text{Sat}) = 0.8V$
 $V_{CE}(\text{Sat}) = 0.2V$

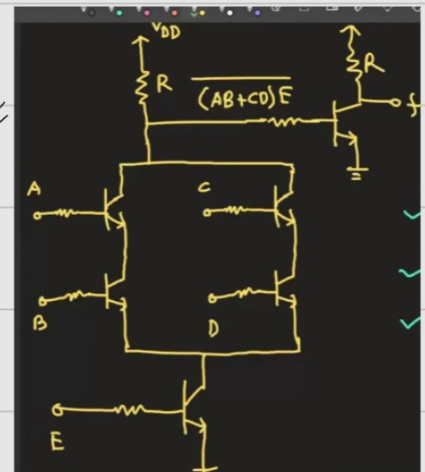
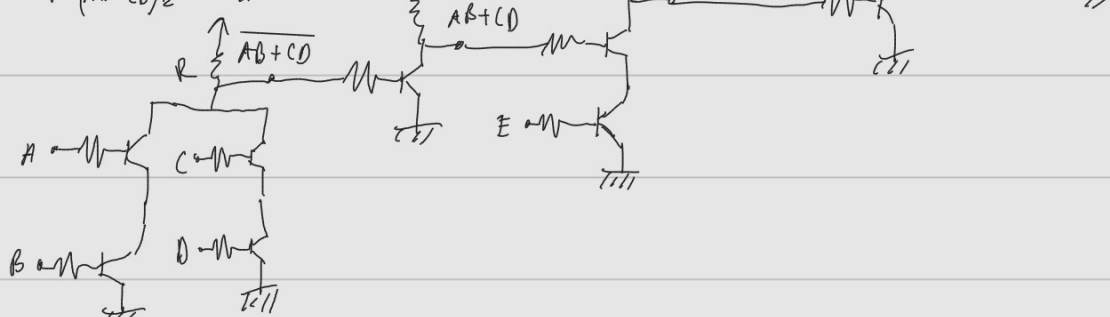
$$10 = I_B \left(\frac{1}{68} + \frac{1}{42} \right) - \frac{10}{68} - \frac{-5}{36} = 0$$

$$V_B = \frac{5}{36}V$$

$$KVL$$

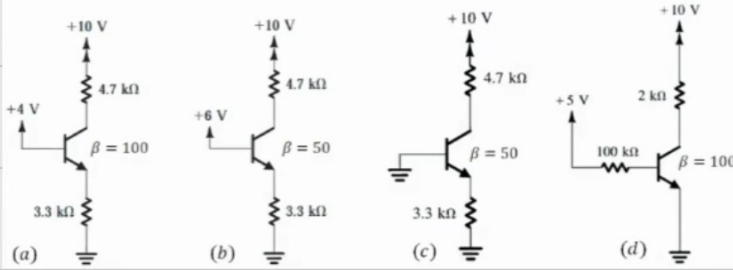
8) Switch-BJT

$$f = (AB + CD)E$$



Problem 3

- Determine and verify the operating region for each of the transistors in the following circuits. Assume $V_{BE, active} = 0.7 \text{ V}$, $V_{BE, sat} = 0.8 \text{ V}$, and $V_{CE, sat} = 0.2 \text{ V}$.



(b) Assuming Active

$$V_{BE} = 0.7 \text{ V}$$

$$\beta = 50$$

$$V_B = 6 \text{ V}$$

$$\therefore V_{BE} = V_B - V_E$$

$$0.7 = 6 - V_E$$

$$\therefore V_E = 5.3 \text{ V}$$

$$I_E = \frac{V_E - 0}{3.3}$$

$$= 1.61 \text{ mA}$$

$$I_E = (\beta + 1) I_B$$

$$\therefore I_B = \frac{1.61}{51} = 0.032 \text{ mA}$$

$$\therefore I_C = \beta I_B = 1.6 \text{ mA}$$

$$\frac{10 - V_C}{4.7} = I_C$$

$$\therefore V_C = 2.48 \text{ V}$$

$$\therefore V_{CE} = 2.48 - 5.3$$

$$< 0.2 \text{ V}$$

$$\therefore \text{Not Active}$$

Assuming Sat

$$V_{BE} = 0.8 \text{ V}$$

$$V_{BE} = V_B - V_E$$

$$\therefore V_E = 5.2 \text{ V}$$

$$I_E = \frac{5.2 - 0}{3.3}$$

$$= 1.58 \text{ mA}$$

$$\# 10 - 0 = 4.7(I_C) + 0.2 + I_E(3.3)$$

$$\therefore I_C = 0.98 \text{ mA}$$

$$\therefore I_E = I_B + I_C$$

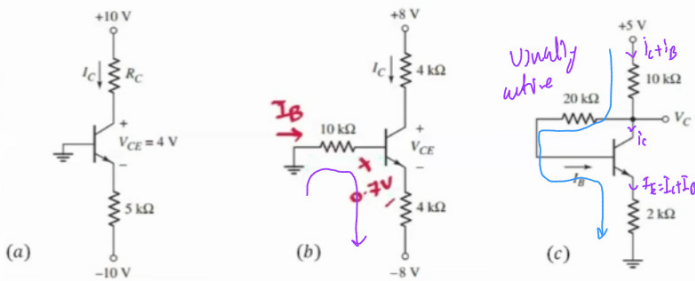
$$I_B = 0.6 \text{ mA}$$

$$\therefore \frac{I_C}{I_B} = \frac{0.98}{0.6} = 1.63 < \beta$$

$$\therefore \text{Sat mode}$$

Problem 4

- For the NPN transistors in the following circuit, $\beta = 75$, $V_{BE, active} = 0.7 \text{ V}$, $V_{BE, sat} = 0.8 \text{ V}$, and $V_{CE, sat} = 0.2 \text{ V}$. Determine the labeled quantities.



(b) Assuming Active

$$V_{BE} = 0.7 \text{ V}, \beta = 75$$

$$\# 0 - (-8) = I_C(10) + V_{BE} + 4I_E$$

$$8 = 10I_C + 0.7 + 4(76I_B)$$

$$\therefore I_B = 0.023 \text{ mA}$$

$$I_C = \beta I_B$$

$$= 75 \times 0.023$$

$$I_C = 1.74 \text{ mA}$$

$$I_E = I_B + I_C = 1.763 \text{ mA}$$

$$\# \frac{V_E - (-8)}{4} = I_E$$

$$V_E = -0.998 \text{ V}$$

$$\# \frac{8 - V_C}{4} = I_C = 1.74$$

$$V_C = 1.04 \text{ V}$$

$$\therefore V_{CE} = 1.04 - (-0.998) > 0.2$$

$$\therefore \text{Active}$$

(c) Assuming Active

$$5 - 0 = 10(I_C + I_B) + 20I_B + 0.7 + 2I_E$$

$$5 = 10 \times (76I_B) + 20I_B + 0.7 + 2(76I_B)$$

$$\therefore I_B = 0.016 \text{ mA}$$

$$I_C = \beta I_B = 0.35 \text{ mA}$$

$$I_E = 0.3546 \text{ mA}$$

$$\# \frac{V_E - 0}{2} = I_E$$

$$V_E = 0.3546 \text{ V}$$

$$\# \frac{5 - V_C}{10} = I_C$$

$$V_C = 1.451 \text{ V}$$

Check

$$V_{CE} = 0.3546 - 1.451$$

$$V_{CE} = 2.02 \text{ V}$$

$$\therefore \text{Active}$$

